Saturday 6:32 PM

ICLR_2027_Submissio...
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Attached is the latest version of my ICRL 2027 submission. I want to focus memory interference and how hippocampal replay may help in alleviating it. I was able to run some experiments last week that show how the replay can help memory transfer from CA3 to CA1 as well (not included in this draft) to that may further assist in preventing interference. However, I am lacking ideas of what happen downstream (CA1 --> subiculum --> EC 5 --> Cortex --> Maybe back to CA3 or EC2/3? Even If I do build this circuitry, I don't know what features I need to build in order to support memory trace encoding and retrieval without

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Yes. I think there is a promising way to extend the paper, but I would avoid trying to build a fully anatomically complete CA3 → CA1 → subiculum → EC → cortex model. The stronger research contribution would be to identify a small set of computational functions that downstream hippocampal–cortical circuitry could perform to reduce interference, and then test those functions in a simplified extension of your existing network.

Your current manuscript already gives you a clean starting point: interference arises because multiple traces are written into the same recurrent CA3 PC–PC matrix, producing synaptic saturation; replay plus anti-hSTDP/homeostasis compresses that representation and extends capacity from roughly five to seven traces. ICLR_2027_Submission (1) +1 Your new CA3→CA1 experiments could provide the next conceptual step: **replay is not only maintaining CA3 memories; it may export them into a representation in which memories can be consolidated, separated, generalized, or enriched.**

After looking across recent neuroscience and continual-learning literature, I would organize the extension around the following architecture.

A useful computational interpretation of the downstream pathway

Rather than thinking of the pathway as simply:

CA3 → CA1 → Subiculum → EC5 → Cortex

I would assign each stage a computational role:

CA3 = episodic/index memory
Fast learning, highly plastic, autoassociative, interference-prone.

CA1 = transformation/readout layer
A sparse learned projection of CA3 replay into another representational space. Importantly, CA1 need not reproduce CA3 exactly.

Deep EC / subiculum = routing/interface layer
Decides where a replayed trace should modify slower downstream representations.

Cortex = slow distributed semantic/schema memory
Learns common structure across many experiences rather than storing every experience verbatim.

There is good biological justification for this broad distinction. The classic hippocampal indexing theory proposes that hippocampus stores an index capable of reinstating distributed cortical representations. PubMed Modern systems-consolidation work likewise emphasizes coordinated hippocampal–neocortical reactivation during sleep, including temporal coupling between hippocampal ripples and cortical slow oscillations/spindles. PubMed Centra... +1

That means you do **not** necessarily need CA1→subiculum→EC5→cortex to be a second sequence-storage network.

I would instead make it a **memory transformation pipeline**.

The most interesting experiment: CA3 replay trains a sparse CA1/cortical code

You now apparently have evidence that CA3 replay can transfer a sequence into CA1. That can become much more important than simply reproducing the same trajectory.

I would test whether CA1 develops a **compressed representation of each memory trace**.

Suppose environment E_i produces a replay vector:

$$x_i^{CA3}(t)$$

During replay, plastic CA3→CA1 synapses produce:

$$z_i^{CA1} = f(W_{CA3\to CA1}x_i)$$

where z_i is deliberately much sparser than x_i.

Then measure overlap:

$$O_{ij} = \frac{|A_i \cap A_j|}{|A_i \cup A_j|}$$

where A_i is the set of CA1 neurons activated by memory i.

This immediately gives you an interference metric.

The important prediction would be:

> CA3 representations become increasingly overlapping as memory load rises, whereas replay-driven CA1 representations remain relatively separated.

That would directly connect your hippocampal model to **representation-based continual learning**.

Recent AI work strongly supports this general strategy. Sparse parameter allocation, task routing and mixture-of-experts reduce catastrophic forgetting by limiting which representations or parameters are shared between tasks. ICLR 2025 theoretical work on MoE continual learning explicitly shows that sparse routing can reduce task interference by allowing experts to specialize. ICLR Proceed... Sparse memory fine-tuning similarly reports that updating only highly relevant memory slots dramatically reduces interference with prior knowledge. ar...

Your biological analogue could be:

CA3 replay → sparse CA1 allocation → selective cortical plasticity.

That is much more interesting than simply adding more neurons downstream.

Memory enrichment

I think this is where your model could become particularly compelling.

Memory enrichment should probably **not** mean "make the original sequence stronger."

Instead, replay could progressively extract **structure shared across experiences**.

Imagine five environments:

$$E_1, E_2, E_3, E_4, E_5$$

CA3 stores them separately.

During offline replay, CA1/cortex sees:

$$E_1, E_3, E_2, E_5, E_1, E_4, \ldots$$

Repeated exposure allows downstream synapses to learn correlations **between memories**, not merely within a memory.

This is very close to the recent generative memory framework from Spens & Burgess. Their model proposes that hippocampal replay trains slower generative representations in entorhinal and cortical systems. The result is that episodic memories increasingly combine unique hippocampal information with schema-like cortical predictions. Nature

This offers a very clean computational definition of **enrichment**:

> Replay enriches a memory by combining its episode-specific hippocampal representation with structure learned across related experiences.

So you could introduce two feature classes.

Unique features

Specific to one trace.

$$u_i$$

Shared/schema features

$$s$$

Each experience becomes:

$$E_i = s + u_i$$

CA3 could encode essentially the complete episode:

$$CA3_i = [s, u_i]$$

But cortex gradually learns:

$$C \approx s$$

Then retrieval reconstructs:

$$\hat{E}_i = C + u_i$$

This is an extremely useful idea for your interference question because **the hippocampus no longer has to store everything**.

It only needs to retain information that distinguishes episode i from the cortical prediction.

That potentially changes your capacity problem dramatically.

Instead of storing:

$$E_i$$

CA3 eventually stores something closer to:

$$E_i - C.$$

That is effectively **residual coding**.

A surprisingly AI-like interpretation

This has a very direct connection to continual learning.

A modern continual-learning system might decompose a task into:

$$\text{shared representation} + \text{task-specific residual.}$$

The shared component is protected and reused while a much smaller task-specific component is allocated to each new task.

Your hippocampal architecture could implement almost exactly this.

Cortex: shared statistical structure
CA3: episodic residual/index
CA1: comparator/transformation
Replay: training signal linking the two

You could describe this as a **biological residual-memory architecture**.

And it gives you an elegant explanation for why replay helps interference:

> Replay progressively transfers redundant information out of the limited-capacity hippocampal representation, leaving the hippocampus responsible primarily for episode-specific information.

That is much stronger conceptually than "replay strengthens memories."

Where CA1 becomes particularly interesting

CA1 is a natural place to calculate something like:

$$\text{mismatch} = \text{hippocampal prediction} - \text{cortical prediction}.$$

There is a large literature treating hippocampal/entorhinal circuitry as predictive rather than purely associative. Your manuscript already cites Liu et al. 2023, where CA1 sequence coding was dissociated from assembly-based associative coding: sequential representations supported predictive transitions while population assemblies supported associative memory. ICLR_2027_Submission (1)

Therefore you could make CA1 representation depend on both:

$$x_{CA3}$$

and

$$x_{EC}.$$

For example:

$$e = x_{CA3} - \hat{x}_{EC}.$$

Large e:

novel episode → strong hippocampal encoding

Small e:

already explained by cortical schema → weak hippocampal storage

That gives you a biologically plausible memory-allocation rule.

It also addresses your question:

> What features should determine whether a memory is encoded without interfering with old memories?

Answer:

novelty relative to existing memory structure.

Replay should probably be selective rather than uniform

There is another important result from the neuroscience literature that fits your work almost perfectly.

Yang et al. showed that waking SWRs selectively tag experiences that are subsequently preferentially replayed during sleep. In other words, replay is not simply random sampling of all experiences. D...

AI researchers have independently arrived at essentially the same idea. Replay-buffer methods increasingly prioritize examples according to information content, representativeness or vulnerability to forgetting rather than replaying everything equally. CORE, for example, adjusts replay allocation according to forgetting and selectively retains representative examples. ar...

You could therefore introduce a replay probability:

$$P(\text{replay } i) \propto \alpha N_i + \beta F_i + \gamma R_i$$

where

- N_i = novelty,
- F_i = estimated vulnerability to forgetting,
- R_i = reward/salience.

That would be a very natural extension of your current model.

And importantly, **you already have an endogenous signal for F_i**.

If a memory becomes difficult to cue/replay, increase its probability of offline replay.

That produces a homeostatic memory scheduler:

$$\text{weak memory} \rightarrow \text{more replay} \rightarrow \text{stronger downstream representation.}$$

You may also want interference to be partly intentional

This is an important conceptual point.

Not all overlap is bad.

The recent engram literature distinguishes **memory interference** from **memory linking**. Experiences close together in time or meaning often recruit overlapping engrams, allowing memories to become associated while retaining their identity. Nature

That suggests a better target than simply minimizing overlap:

$$\text{maximize useful overlap} - \text{minimize destructive overlap.}$$

For highly similar environments:

CA1/cortex should share neurons.

For unrelated environments:

CA1/cortex should separate them.

So define similarity:

$$S_{ij} = \mathrm{similarity}(E_i, E_j).$$

Then an ideal representation would have:

$$\mathrm{Overlap}(z_i, z_j) \propto S_{ij}.$$

That is a beautiful experiment because it moves you from **capacity** to **structured continual memory**.

Liao et al. 2024 is especially relevant here because they showed that inhibitory plasticity can support replay generalization rather than merely preserving exact memories. Nature

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I would therefore build three experiments, not an entire brain

If I were choosing the next simulations for this paper, I would prioritize these.

Experiment 1 — Replay-mediated CA3 → CA1 transfer

You are already doing this.

Learn environments sequentially and measure whether CA1 develops trace-specific representations during offline replay.

Compare:

No replay vs replay

Measure:

- CA1 sequence fidelity
- recall after CA3 degradation
- CA3/CA1 overlap across environments
- synaptic sparsity
- number of memories retained.

Your key result could be:

> Replay creates downstream memory representations before CA3 capacity is exhausted.

Experiment 2 — Pattern separation downstream

Give CA1 a competitive sparse activation rule.

For example:

$$k = 5\% - 10\%$$

of neurons active for each event.

Hebbian plasticity:

$$\Delta w_{ij} = \eta x_i y_j$$

with homeostatic normalization.

Then ask:

$$\text{Does replay generate increasingly orthogonal CA1 engrams?}$$

Compare:

$$\text{Overlap}_{CA3}$$

against

$$\text{Overlap}_{CA1}.$$

If CA1 remains sparse while CA3 becomes saturated, you have a direct mechanism for interference reduction.

Experiment 3 — Schema extraction / enrichment

This might ultimately be the most interesting.

Create environments with controlled shared structure.

For example:

A: 1→2→3→4→5

B: 1→2→3→6→7

C: 1→2→3→8→9

All contain a common segment.

Repeated replay should cause cortex to learn:

$$1 \to 2 \to 3$$

as a shared structure.

CA3 only needs to encode the branch:

$$4, 5$$

versus

$$6, 7$$

versus

$$8, 9.$$

Now test whether consolidation increases the number of memories that can coexist.

This would give you an operational definition of **memory enrichment**:

> The system extracts structure common to multiple episodic traces and incorporates it into a reusable downstream representation.

That connects naturally to schema formation, semanticization and generalization. Systems-consolidation theories specifically predict that repeated experience can transform detailed hippocampal representations into more gist-like/schema-like extrahippocampal representations. PubMed +1

A fourth experiment could be extremely powerful

After a memory has been consolidated downstream, deliberately weaken or delete part of its CA3 representation.

Then cue the memory.

Ask whether:

$$CA1/Cortex \to EC \to CA3$$

can reconstruct the missing hippocampal sequence.

You do not need to reproduce the anatomy perfectly.

Computationally you can simply give cortex/CA1 feedback connections into CA3.

Then compare:

Early memory

$$CA3 \rightarrow CA1/Cortex$$

with

consolidated memory

$$Cortex \leftrightarrow CA3.$$

That gives you a closed loop.

And there is biological support for treating EC as more than a one-way relay: deep medial entorhinal outputs can send information both toward telencephalic targets and back toward CA1, suggesting EC participates in coordinating hippocampal–cortical interactions rather than simply forwarding hippocampal output. PubMed

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The larger paper narrative could become very strong

Your current paper asks:

> **How can CA3 continue learning without destroying previously learned memory traces?**

The current answer is:

> replay + anti-hSTDP + homeostasis compresses recurrent connectivity and delays interference. ICLR_2027_Submission (1)

The next result could change the question slightly:

> **Perhaps CA3 is not supposed to retain an unlimited number of fully specified memories.**

Instead:

$$\boxed{\text{CA3 episodic encoding} \rightarrow \text{Replay} \rightarrow \text{CA1 transformation} \rightarrow \text{Cortical consolic}}$$

and then

$$\boxed{\text{Cortical structure} + \text{hippocampal residual} \rightarrow \text{memory reconstruction.}}$$

That elegantly explains why your existing compression mechanism only gets you from approximately five to seven traces rather than unlimited capacity. ICLR_2027_Submission (1)

Perhaps that finite capacity is not a defect of the model.

It may be pointing toward the biological necessity of systems consolidation.

The AI connection I would emphasize

The continual-learning literature broadly uses four families of solutions:

replay, **regularization/protection**, **sparse or modular architectures**, and **representation separation**. Recent CL surveys still organize the field around closely related categories. DOI +1

Your system could eventually contain biological analogues of all four:

Continual-learning idea	Your biological mechanism
Experience replay	hippocampal replay
Parameter pruning/reset	synaptic homeostasis
Regularization	anti-hSTDP
Sparse parameter allocation	sparse CA1/cortical engrams
Mixture-of-experts routing	EC/subiculum routing
Generative replay	cortical reconstruction driven by hippocampal index
Latent replay	CA1 rather than raw sensory replay
Knowledge distillation	hippocampus → cortex consolidation

That is a very compelling neuroscience/AI bridge.

The recent Spens & Burgess model is probably the single paper I would read next in depth. Their central idea—that hippocampal replay trains slower generative cortical

representations—is remarkably close to where your experiments appear to be heading.
Nature

My preferred direction for your ICLR submission would therefore be **not to model every downstream anatomical structure**. I would add a simplified CA1 plus "cortical memory" layer and show three things: replay transfers traces out of CA3, sparse downstream representations reduce interference, and repeated replay extracts common structure across related memories. If you can demonstrate even the first two convincingly before the deadline, it would substantially strengthen the paper's continual-learning story; the schema/enrichment experiment could then become either the strongest final result or the basis of the next paper.

I have few issues with the way that you describe experiment 1. CA1 does not have recurrent PC --> PC connections, so all CA1 does from PC spiking pattern is passing on CA3 output with minimal processing. If CA3 replay is degraded, then CA1 replay will degrade as well.. Unless CA3 --> CA1 connections (Schaffer collaterals) have a plasticity rule or other features that allow the CA1 replay to be somehow preserved... which I am not clear on. I did check and identified sSTDP during learning and some sort of hSTDP or Anti-hSTDP during replay to preserve the replay output without too disruption. Actually, I

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I agree with your criticism. My earlier framing gave CA1 too much autonomy. In the architecture you are actually modeling, **CA1 is not another CA3-like attractor network**. Without recurrent CA1 pyramidal-to-pyramidal connectivity, it cannot independently regenerate a degraded sequence. Its output during replay will depend heavily on the structure of the incoming CA3 activity and on how the Schaffer collateral weights transform that activity.

That changes what Experiment 1 should ask.

Your manuscript currently treats replay as a property generated by recurrent CA3 connectivity and shows that this representation progressively degrades as interference accumulates. ICLR_2027_Submission (1) If you now add CA1, the scientifically interesting question is not whether CA1 can "save" a corrupted CA3 replay. It is:

> **Can plasticity at CA3→CA1 synapses create a stable downstream readout of replay while CA3 representations are still intact, and can that readout resist interference as additional CA3 memories are learned?**

That is a much more defensible experiment.

The Schaffer collateral literature actually gives you more room than just classical hSTDP

Your recollection of Bi and Poo is directionally right regarding classic Hebbian STDP, but there is an important qualification. Bi & Poo's canonical experiments used cultured hippocampal neurons, rather than an intact CA3→CA1 Schaffer collateral circuit. A later review of hippocampal STDP explicitly notes that the original hippocampal STDP studies came from dissociated or cultured preparations and that obtaining a single universal STDP rule in acute hippocampal slices has been difficult. PubMed

More directly relevant is Wittenberg & Wang's work specifically at **CA3–CA1 synapses**. They found that the plasticity rule is highly state- and activity-dependent rather than a simple fixed pre-before-post = LTP / post-before-pre = LTD window. Potentiation required postsynaptic bursting at sufficiently high frequencies, while depression could emerge under different repetition regimes. They could obtain LTP-only, bidirectional, or LTD-only timing curves depending on the broader activity pattern. PubMed

That matters a great deal for your model.

It gives you a biologically grounded reason to use **different effective plasticity regimes during exploration and replay**, without claiming that Schaffer collaterals literally switch between a textbook hSTDP rule and your anti-hSTDP rule.

In other words, I would be careful about saying:

> "Schaffer collateral synapses use anti-Hebbian STDP during replay."

I don't think the evidence supports that statement.

But you can reasonably investigate:

> "whether a replay-state-dependent synaptic plasticity rule with net anti-Hebbian/depressing characteristics stabilizes CA3→CA1 transmission."

That is a computational hypothesis rather than a claim about established CA1 physiology.

And there is another especially relevant experimental result. During SWRs, replayed CA3 and CA1 spike patterns can induce plasticity at CA3→CA1 synapses. Sadowski et al.

showed that replay-like spike trains recorded simultaneously from CA3 and CA1 could induce LTP at Schaffer collateral synapses, but crucially only in the presence of the SWR-associated depolarization state; the timing relative to the SWR also mattered. PubMed

So there is quite strong support for:

$$\text{replay state} \Rightarrow \text{Schaffer collateral plasticity.}$$

What is much less established is the exact learning kernel operating during that state.

This suggests a better interpretation of your anti-hSTDP finding

Suppose you observe experimentally in your simulations:

sSTDP during exploration + anti-hSTDP during replay

preserves CA1 replay better than

sSTDP during exploration + hSTDP during replay.

I wouldn't interpret that as evidence that biological Schaffer collaterals literally undergo anti-hSTDP.

Instead, I would interpret anti-hSTDP as implementing a **normalizing/decorrelating function**.

Consider what ordinary Hebbian plasticity does during replay. If CA3 repeatedly produces:

$$A \to B \to C \to D$$

and CA1 follows:

$$a \to b \to c \to d,$$

then replay with Hebbian STDP keeps strengthening precisely those already strongly correlated CA3→CA1 pairs.

After many replay events:

$$w_{Aa}, w_{Bb}, w_{Cc}, w_{Dd} \uparrow .$$

That could progressively saturate Schaffer collaterals just as sSTDP saturates your recurrent CA3 matrix.

Your current CA3 findings already demonstrate the general problem: repeated associative strengthening produces an increasingly saturated weight distribution and eventually destroys useful representational structure. ICLR_2027_Submission (1)

Anti-hSTDP during replay might instead counterbalance that process:

$$\text{strongly correlated pathway} \rightarrow \text{slight weakening}$$

while preserving enough structure for transmission.

So its role would be less:

"learn the replay sequence"

and more:

"prevent replay from overtraining the readout."

That distinction could become very important in the paper.

I think there are actually three Schaffer-collateral hypotheses worth testing

You already have much of the machinery to test them.

1. Plasticity off during replay

This should be your baseline.

During exploration:

$$CA3 \xrightarrow{sSTDP} CA1$$

During replay:

$$\Delta W_{CA3 \rightarrow CA1} = 0.$$

Question:

Does the original learned projection remain usable as more environments are learned?

If it fails, you establish that **some form of offline Schaffer-collateral plasticity is beneficial**.

2. Hebbian plasticity during replay

$$CA3 \xrightarrow{hSTDP} CA1$$

This is arguably the easiest condition to motivate biologically because replay-associated Schaffer collateral potentiation has experimental support. PubMed +1

But your hypothesis could be that continual replay progressively causes:

$$\text{SC saturation} \rightarrow \text{CA1 interference.}$$

That would mirror your CA3 result.

3. Anti-Hebbian / homeostatic replay plasticity

$$CA3 \xrightarrow{anti\text{-}hSTDP} CA1.$$

Now ask whether it:

- preserves replay fidelity,
- keeps CA1 firing stable,
- keeps the Schaffer collateral weight distribution sparse,
- reduces cross-memory activation,
- and delays interference.

This could be extremely interesting even if anti-hSTDP is framed explicitly as a **candidate computational approximation of sleep-state synaptic normalization**, rather than established Schaffer-collateral physiology.

You would essentially have found the same computational principle operating at two successive stages:

$$\boxed{\text{Replay} + \text{potentiation alone} \rightarrow \text{saturation}}$$

versus

$$\boxed{\text{Replay} + \text{selective weakening/homeostasis} \rightarrow \text{stable memory transmission.}}$$

That could become a central theme of the paper.

CA1 can still perform meaningful processing without recurrent excitation

There's another subtle point worth making.

CA1 does not have to simply copy CA3 one-for-one.

The absence of strong recurrent pyramidal excitation means CA1 isn't an autoassociative completion network like CA3, but **feedforward connectivity + inhibition + intrinsic excitability + Schaffer collateral weighting can still transform the population representation**.

Classic simultaneous CA3/CA1 recordings during sharp waves found that only sufficiently synchronous CA3 population events strongly recruit CA1, and individual subsets of CA3 cells preferentially predicted firing of particular CA1 groups. In other words, CA1 output is not merely a passive duplicate of every CA3 spike. PubMed

So conceptually you can write:

$$y_{CA1} = f(W_{SC} x_{CA3}, I_{CA1}),$$

rather than:

$$y_{CA1} = x_{CA3}.$$

The important processing terms can be very modest:

1. **weighted Schaffer collateral projection,**
2. **feedforward/feedback inhibition,**
3. **thresholding**, and perhaps
4. **plasticity at Schaffer collaterals.**

You don't need CA1 recurrence to get a cleaner or sparser representation.

What it cannot do is reconstruct information that is **no longer present anywhere in its input or weights**.

And that gets us to your degraded-CA3 objection.

There is one circumstance where CA1 could partially survive degraded CA3 replay

Suppose CA3 replay becomes degraded because some of the sequence-generating recurrent connections have weakened, but enough CA3 cells still participate.

For example, originally:

$$A - B - C - D - E - F$$

and later CA3 produces:

$$A - \_ - C - D - \_ - F.$$

If CA3→CA1 connectivity has undergone associative learning, a CA1 neuron can receive convergent input from multiple CA3 cells.

Then:

$$A + C + D$$

could still be sufficient to activate a CA1 ensemble even though the exact CA3 sequence is incomplete.

This isn't recurrent pattern completion in CA1.

It is **redundant feedforward decoding**.

Mathematically:

$$y_j = H\left(\sum_i w_{ij} x_i - \theta_j\right).$$

If a CA1 neuron originally received strong input from several CA3 neurons associated with one portion of a trace, loss of one input doesn't necessarily eliminate its activation.

So CA1 could be **more robust than the raw CA3 raster**, but only to partial/noisy degradation, not complete loss of the CA3 engram.

That would actually be worth measuring.

Instead of asking:

> Does CA1 replay survive when CA3 replay disappears?

ask:

> **How does CA1 decoding fidelity change as CA3 replay fidelity progressively degrades?**

That gives you a transfer function:

$$F_{CA1} = g(F_{CA3}).$$

Three possibilities are interesting:

$$F_{CA1} \approx F_{CA3}$$

pass-through,

$$F_{CA1} < F_{CA3}$$

error amplification,

or

$$F_{CA1} > F_{CA3}$$

error-tolerant decoding.

That last result would be genuinely interesting.

Your anti-hSTDP result suggests an especially good metric

Don't only count CA1 replays.

Measure **CA3→CA1 representational fidelity**.

For memory m, take the expected CA1 sequence learned during exploration:

$$S_m^{learn}.$$

For each subsequent CA3 replay, measure:

$$F_m = corr(S_m^{replay}, S_m^{learn})$$

or use the same regression/order-based measure you're already using.

Then track:

$$F_m(n)$$

as the network learns memory traces $1, \ldots, n$.

Compare:

SC plasticity	CA1 fidelity	SC saturation	Cross-trace interference
None	?	low	?
hSTDP	?	probably higher	?
anti-hSTDP	?	potentially lower	?
hSTDP + homeostasis	?	lower	?
anti-hSTDP + homeostasis	?	perhaps lowest	?

That experiment would fit your existing paper extremely well because you're already analyzing synaptic distributions and interference in essentially this way for recurrent CA3.

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One biological direction may be safer than calling it anti-Hebbian STDP

You might eventually want to test a rule like:

$$\Delta w = \Delta w_{\text{timing}} - \lambda w$$

or

$$\Delta w = \Delta w_{\text{Hebbian}} - \lambda r_{\text{pre}} r_{\text{post}}.$$

That gives you **Hebbian sequence transfer plus activity-dependent normalization**.

Biologically, this is probably easier to defend than pure anti-hSTDP.

The Wittenberg & Wang result is useful precisely because it demonstrates that CA3→CA1 plasticity is not described by one fixed timing curve. Depending on firing pattern, they observed potentiation-only, bidirectional, or depression-dominated plasticity. PubMed

And recent sleep work reinforces the broader idea that replay is accompanied by mechanisms actively counterbalancing excitation. For example, Karaba et al. found a sleep-associated inhibitory circuit that preferentially suppresses recently potentiated CA1 assemblies and returns reactivation toward baseline; disrupting that balancing mechanism increased CA1 synchrony and impaired consolidation. PubMed

That's not Schaffer-collateral anti-hSTDP.

But computationally it supports the concept:

$$\text{reactivation} + \text{down-regulation}$$

rather than

$$\text{reactivation} + \text{unbounded Hebbian strengthening.}$$

That seems strikingly consistent with what your simulations are telling you.

So I would revise the CA3→CA1 story substantially

I would no longer frame it as:

> **Replay transfers memories from CA3 into an independent CA1 memory store.**

Instead:

> **Replay trains and maintains a plastic feedforward hippocampal readout through the Schaffer collateral pathway. The stability of this downstream representation**

> **depends on how replay-driven plasticity balances associative strengthening against synaptic saturation.**

That's both more anatomically defensible and more tightly connected to your existing interference result.

And I think your observation that **anti-hSTDP preserves the CA1 replay output better than hSTDP may actually be quite important**, even though you'll need to be restrained about its biological interpretation.

The interesting claim isn't necessarily:

> anti-Hebbian STDP occurs at Schaffer collaterals during sleep.

It could be:

> **A purely Hebbian replay rule is computationally unstable under continual learning, whereas a replay-dependent weakening/normalization component stabilizes hippocampal output.**

Then neuroscience gives you several possible biological implementations of that computational requirement—state-dependent LTD, heterosynaptic plasticity, synaptic scaling, inhibitory regulation, or some combination—without requiring you to prove that your precise anti-hSTDP kernel exists in vivo.

That, to me, is a much stronger and safer direction for the model.

Please provide reference for Wittenberg & Wang

The reference I was referring to is:

Wittenberg, G. M., & Wang, S. S.-H. (2006). Malleability of spike-timing-dependent plasticity at the CA3–CA1 synapse. *Journal of Neuroscience, 26*(24), 6610–6617. https://doi.org/10.1523/JNEUROSCI.5388-05.2006 . PubMed +1

This is particularly relevant to your Schaffer-collateral question because they studied plasticity specifically at the **CA3→CA1 synapse** and showed that the STDP rule was not fixed. The resulting LTP/LTD depended not only on pre/post spike timing, but also on factors such as firing frequency, postsynaptic bursting, and the number/repetition of pairings. Under different stimulation conditions they obtained LTP-dominated, bidirectional, or LTD-dominated timing relationships. The Journal of ...

So I think it is a useful citation for your model—not as evidence for **anti-hSTDP at Schaffer collaterals**, because it does not show that—but as evidence that CA3→CA1 plasticity is **state/activity dependent and considerably more flexible than a single canonical Hebbian STDP curve**.

This distinction could help justify testing multiple replay-phase plasticity rules in your model.

Dembitskaya et al 2026...
PDF

I am curious whether a recent paper (That is citing my prior work - see ref 53) can shed more light on my research for ICLR - See attached.

Yes. This paper is quite relevant to your ICLR direction, but probably **not for the reason implied by their citation of your anti-hSTDP paper**. The most useful contribution for you is that Dembitskaya et al. challenge the assumption that a fixed pairwise STDP kernel is an adequate description of plasticity under naturalistic replay. That could give you a more biologically defensible way to formulate your next experiments.

Their citation of your paper is very specific: after noting that anti-Hebbian STDP exists in several circuits, they cite your work as a theoretical example in which anti-Hebbian STDP can stabilize hippocampal replay dynamics. Dembitskaya et al 2026 - Irregu... Their actual experiments, however, are at **cortico-striatal synapses**, so they do not provide direct evidence that Schaffer collaterals use your anti-hSTDP rule.

The important insight for your model: plasticity may depend on the *statistics of replay*, not just pairwise Δt

Their central result is striking. With regular low-frequency spike pairs, they observe a conventional timing-sensitive plasticity window. But when the same average activity becomes irregular, LTD is reduced and the LTP window becomes substantially broader; at higher rates, timing dependence nearly disappears and LTP dominates.
Dembitskaya et al 2026 - Irregu...

The mechanism in their calcium model is quite intuitive. Calcium transients generated by successive spikes overlap when spikes cluster in time. Once calcium crosses the potentiation threshold for sufficiently long periods, the sign of plasticity becomes dominated by the recent **pattern of activity**, rather than simply:

$$\Delta t = t_{\text{post}} - t_{\text{pre}}.$$

This matters enormously for replay because replay is exactly the kind of activity for which a simple pairwise STDP interpretation becomes questionable: replay is compressed, irregular, and bursty.

Their most interesting result is that **bursts account for most of the plasticity induced by irregular firing**. At 3 spikes/s, roughly 60% of spikes occur in bursts, but using only those bursts reproduced approximately the same LTP as the entire irregular spike train—even though many isolated events were removed. Dembitskaya et al 2026 - Irregu...

That suggests a different conceptual model for your CA3→CA1 experiments:

$$\boxed{\text{Replay event statistics} \rightarrow \text{Schaffer collateral plasticity}}$$

rather than:

$$\boxed{\Delta t \rightarrow \text{fixed STDP rule}}.$$

That could be quite important.

It also changes how I would interpret your anti-hSTDP result

Right now you have something like:

online exploration

$$CA3 \rightarrow CA1: \ sSTDP$$

and then during replay you have compared:

$$hSTDP$$

against

$$anti\text{-}hSTDP.$$

You observe that anti-hSTDP seems better at preserving the CA1 replay output.

Initially, this looks awkward biologically because Schaffer collateral literature tends to provide more evidence for LTP/Hebbian plasticity during replay than for an explicit anti-Hebbian kernel.

Dembitskaya et al. give you another possibility:

> **The problem might be the assumption that replay plasticity should use a conventional pair-STDP kernel at all.**

Their results show that changing spike statistics while leaving the synaptic machinery unchanged can radically change the effective learning rule. At sufficiently irregular/bursty activity, the temporal contingency of conventional STDP becomes much less important.

Dembitskaya et al 2026 - Irregu…

That means you might eventually replace the question

> hSTDP or anti-hSTDP during replay?

with the more interesting question

> **What effective plasticity rule emerges from compressed replay firing statistics?**

That is substantially richer scientifically.

And this links directly to Wittenberg & Wang

I find it particularly interesting that this new paper cites Wittenberg & Wang as reference 32. Dembitskaya et al 2026 - Irregu…

As we discussed, Wittenberg & Wang showed that plasticity at the **CA3→CA1 synapse** depends strongly on more than pre/post order: frequency, bursting, and repetition substantially alter the rule.

Dembitskaya et al. now provide a much more mechanistic explanation of exactly that general phenomenon:

$$\text{timing} + \text{frequency} + \text{burst structure} \rightarrow \text{effective plasticity.}$$

Their calcium-based rule is:

$$\tau_\rho \dot{\rho} = \ldots + \gamma_p(1-\rho)\Theta[c(t)-\theta_p] - \gamma_d \rho \Theta[c(t)-\theta_d] + \ldots$$

with calcium:

$$c(t) = \sum_i C_{\text{pre}} e^{-(t-t_i-D)/\tau_{Ca}} + \sum_j C_{\text{post}} e^{-(t-t_j)/\tau_{Ca}}.$$

Dembitskaya et al 2026 - Irregu…

This has an appealing property for your purposes:

you don't explicitly tell the synapse whether to use Hebbian or anti-Hebbian STDP.

The plasticity emerges from the temporal integration of the actual pre/post spike trains.

That would be much more biologically interesting than assigning a hand-selected replay kernel.

There is also a direct Schaffer-collateral result buried in this paper

This is probably the most useful paragraph for your immediate CA3→CA1 question.

In their discussion they specifically mention experiments at **Schaffer collateral→CA1 synapses** showing that six pairings can induce NMDAR-dependent LTP, while pairing a single presynaptic spike with a four-spike postsynaptic burst induces LTP through somewhat different signaling mechanisms. Dembitskaya et al 2026 - Irregu...

They cite Cepeda-Prado et al. 2022 for this:

Cepeda-Prado et al. (2022), "Calcium-Permeable AMPA Receptors Mediate Timing-Dependent LTP Elicited by Low Repeat Coincident Pre- and Postsynaptic Activity at Schaffer Collateral-CA1 Synapses." Dembitskaya et al 2026 - Irregu...

That reinforces something important for your replay modeling:

> A relatively small number of appropriately patterned CA3/CA1 coactivations may be sufficient to modify Schaffer collateral connectivity.

So your 25-second replay periods could potentially produce substantial CA3→CA1 reorganization without requiring hundreds of identical replay events.

I would therefore consider a new experiment before building cortex

This might actually be more valuable for the current ICLR paper than immediately extending CA1→subiculum→EC→cortex.

Take the CA3 and CA1 spike trains you already generate.

For every CA3→CA1 synapse, compare three plasticity models during replay:

A. Your current pairwise hSTDP

$$\Delta w = f(\Delta t)$$

B. Your anti-hSTDP

$$\Delta w = -f(\Delta t)$$

C. Activity-dependent/calcium plasticity

$$\Delta w = f\left(c(t), \theta_{\mathrm{LTD}}, \theta_{\mathrm{LTP}}\right).$$

Then don't tune C to make replay work.

Use parameters inspired by experimentally measured CA3→CA1 calcium/plasticity literature where feasible.

The question becomes:

> **How does naturalistic replay activity itself reshape the CA3→CA1 synaptic representation?**

You could then characterize replay events according to:

- instantaneous CA3 firing rate,
- burst probability,
- CA3–CA1 spike timing,
- replay compression,
- duration,
- synchrony.

And ask which properties predict:

$$LTP, \quad LTD, \quad \text{or no change.}$$

That would be a very nice mechanistic result.

There is a particularly interesting prediction for your model

Your replay events are highly temporally compressed.

Therefore they contain many short interspike intervals.

Dembitskaya et al.'s mechanism predicts:

$$\text{greater temporal clustering} \rightarrow \text{calcium accumulation} \rightarrow \text{greater probability of}$$

Their Figure 2 makes this visually obvious: irregular firing shifts the plasticity curve upward toward LTP as frequency rises, eventually making potentiation almost independent of Δt.

This creates an interesting possibility.

During ordinary exploration

Your place-cell sequences are theta-modulated and relatively slow:

$$A \to B \to C \to D$$

over seconds.

Here, timing-sensitive STDP can encode the relationship.

During replay

The same sequence becomes:

$$A \to B \to C \to D$$

over tens/hundreds of milliseconds.

Now calcium summation becomes much stronger.

So **the same synapse could effectively operate differently during learning and replay without changing its molecular learning rule at all.**

That is quite an elegant hypothesis:

$$\boxed{\text{same synaptic mechanism} + \text{different activity regime} = \text{different effective plas}}$$

That would be much preferable biologically to simply switching:

$$sSTDP \to anti\text{-}hSTDP.$$

But there is an apparent problem—and it could become an interesting result

Dembitskaya et al.'s naturalistic rule tends toward **LTP**, not LTD.

Your simulations seem to tell you:

$$\text{too much potentiation} \to \text{synaptic saturation} \to \text{interference.}$$

That is precisely what your current ICLR manuscript demonstrates in CA3: recurrent sSTDP progressively pushes more synapses into stronger weight ranges, replay degrades beyond four/five memories, and compression restores some capacity.

ICLR_2027_Submission (1)

So if replay at CA3→CA1 produces burst-dependent LTP, you may simply move the saturation problem downstream:

$$CA3 \text{ saturation} \quad \longrightarrow \quad SC \text{ saturation}.$$

That would suggest another mechanism is required.

And *that* mechanism may be your central story.

Replay-induced potentiation + synaptic competition

Instead of making anti-hSTDP itself the biological mechanism, try:

$$\text{Replay-driven LTP} + \text{homeostatic weakening}.$$

For example:

$$\Delta w_{ij} = \Delta w_{ij}^{Ca} - \lambda w_{ij}.$$

Or normalize incoming SC weights for every CA1 neuron:

$$w_{ij} \leftarrow \frac{w_{ij}}{\sum_i w_{ij}} C.$$

Now replay selects the synapses that matter:

$$\text{strongly replay-correlated SC synapses} \rightarrow LTP$$

while homeostasis removes less useful ones:

$$\text{weak/nonparticipating SC synapses} \rightarrow LTD/reset.$$

That's quite close conceptually to what is already happening in your CA3 model.

You could therefore arrive at a much broader principle:

$$\boxed{\text{Replay selects} + \text{homeostasis compresses} = \text{interference-resistant memory}}$$

And this principle could operate both within CA3 and during hippocampal output.

Your current paper already provides strong evidence for the second half: anti-hSTDP plus homeostatic compression disproportionately removes moderately potentiated synapses while leaving replay intact. ICLR_2027_Submission (1)

This also helps your AI framing

There is an interesting analogy to continual learning.

Experience replay in AI often suffers from a similar conceptual issue: simply replaying old samples and applying the same gradient updates can still produce interference. Brain-inspired replay research has shown that spontaneous/sleep-like replay can mitigate catastrophic forgetting, including by producing sparser class representations. Nature +1 More recent work also emphasizes sparse updates as a way of allowing learning while minimizing interference with previously stored knowledge. ar...

What Dembitskaya adds to your story is a potential biological counterpart:

> **Not every replayed spike is an equal training sample. Temporal clustering determines which replay events produce large synaptic updates.**

That's conceptually similar to **importance-weighted replay**.

A burst:

$$\text{small number of events} \to \text{large learning signal.}$$

An isolated spike:

$$\text{event} \to \text{little synaptic modification.}$$

This suggests that **replay structure itself acts as a synaptic importance signal.**

I think that is a really interesting idea for your paper.

And it suggests an experiment on memory interference that I like even more

Right now you classify an entire replay as valid/not valid.

Instead, characterize each replay's **burst structure**.

For replay r, compute perhaps:

$$B_r = \frac{N_{\text{spikes in bursts}}}{N_{\text{spikes total}}}.$$

Then track whether replays with high B_r:

1. produce larger CA3→CA1 synaptic changes,
2. establish stronger CA1 representations,

3. survive subsequent memory learning better.

You could eventually test:

$$P(\text{memory survives}) = f(B_r, N_r, \text{synchrony}_r).$$

That gives you an endogenous mechanism for determining **which memories consolidate**.

Memories producing strong/coherent replay:

$$\rightarrow \text{bursty CA3/CA1 activation} \rightarrow \text{strong plasticity} \rightarrow \text{consolidation.}$$

Weak/noisy memories:

$$\rightarrow \text{isolated activity} \rightarrow \text{little plasticity} \rightarrow \text{eventual forgetting.}$$

That starts addressing the "memory enrichment" problem you raised initially without adding five more anatomical regions.

Where I think this could take the ICLR paper

The current manuscript is essentially:

$$\boxed{\text{Sequential learning} \rightarrow \text{synaptic saturation} \rightarrow \text{interference}}$$

and

$$\boxed{\text{Replay-driven compression} \rightarrow \text{reduced interference.}}$$

ICLR_2027_Submission (1)

The CA1 extension could make the story:

$$\text{CA3 encoding}$$

↓

$$\text{compressed replay}$$

↓

$$\text{burst-dependent plasticity at CA3}\rightarrow\text{CA1}$$

↓

$$\text{selective downstream transfer}$$

↓

$$\text{homeostatic competition}$$

$$\downarrow$$

$$\text{interference-resistant representation.}$$

That is more mechanistically satisfying than simply "CA3 replay trains CA1."

And importantly, the Dembitskaya paper gives you a **very timely experimental justification** for considering naturalistic spike statistics rather than fixed STDP windows. Their explicit discussion of Wittenberg & Wang, Schaffer collateral plasticity, low-repeat learning, irregular firing and bursts gives you a credible chain of literature connecting their cortico-striatal experiment to the particular hypothesis you're testing at CA3→CA1.

Dembitskaya et al 2026 - Irregu...

I would therefore put **calcium-/burst-dependent Schaffer collateral plasticity quite high on the list of experiments worth trying next**. Even a relatively small result showing that the plasticity induced by your actual replay spike trains differs fundamentally from what is predicted by pairwise hSTDP would add a novel mechanistic dimension to the ICLR paper—and would also make your use of anti-hSTDP easier to interpret: it becomes one computational approximation to an activity-dependent stabilizing process rather than a claim that SC synapses literally implement your anti-Hebbian kernel.

One caveat matters: Dembitskaya et al.'s primary experimental synapse is corticostriatal, not CA3→CA1, so I would use it to motivate **the importance of naturalistic spike statistics**, not as direct evidence for the polarity of Schaffer-collateral plasticity. The Schaffer-collateral bridge should instead lean on Wittenberg & Wang, Cepeda-Prado et al., and Sadowski et al., all of which Dembitskaya themselves place in this broader context.

Dembitskaya et al 2026 - Irregu...

I think that Schaffer collaterals are not the issue for memory interference. I don't agree with your analysis. I think there need to be something else. The idea I was thinking off is that CA3 has limited storage capacity for new memory traces (like a cache in computer architecture). Replay from CA3 --> CA1 --> Subiculum --> EC can establish a memory transfer to other parts of the brain where the memory will be preserved. I don't think I even need to make my model very bio-feasible. It's an ICLR submission, not a Nature submission. My issue is that once the replay leaves CA1, I don't know much about the downstream impact of that

Show more

I think your **CA3-as-cache** framing is much closer to the problem you actually want to solve. In that framing, Schaffer-collateral interference is secondary. The key question becomes: **what does downstream circuitry do with replay so that a trace can eventually stop occupying scarce CA3 recurrent capacity?**

That is a very different problem from asking whether CA1 can rescue a degraded replay.

Your current manuscript already gives you the motivation: CA3 recurrent storage saturates under continual learning, and replay-driven compression delays but does not eliminate that capacity limit. ICLR_2027_Submission (1) A natural extension is therefore:

$$\text{fast hippocampal store} \rightarrow \text{offline replay} \rightarrow \text{downstream consolidation} \rightarrow \text{CA3}$$

That is also broadly consistent with systems-consolidation theory, where hippocampal replay is thought to drive distributed extra-hippocampal representations that can eventually support memory with reduced hippocampal dependence. PubMed +1

Subiculum may actually give you a very useful intermediate computation

The paper you were remembering is:

Pandey, A. & Sikdar, S. K. (2014). "Depression biased non-Hebbian spike-timing-dependent synaptic plasticity in the rat subiculum." ***Journal of Physiology*** **592:3537–3557.**

They studied proximal excitatory inputs onto subicular pyramidal neurons. With a presynaptic EPSP followed by a postsynaptic burst, they obtained LTD; reversing the order produced LTP. Importantly, this occurred in both regular-firing and weak-burst-firing subicular neurons, although the underlying mechanisms differed between the two classes. Physiological S...

That is indeed the experimental basis you used for the anti-Hebbian rule in your earlier replay paper. Wiley Online Li...

But I think the **Böhm et al. 2015** paper may be even more useful for deciding what SUB should do computationally:

Böhm et al. (2015), "Functional Diversity of Subicular Principal Cells during Hippocampal Ripples." J Neurosci 35:13608–13618. PubMed

They find that the two subicular principal-cell classes participate very differently during hippocampal ripples. **Bursting cells are preferentially recruited during ripples**, whereas

regular-firing cells participate less strongly. They interpret bursting cells as a preferential channel for transmitting ripple-associated hippocampal information downstream.
PubMed +1

That is almost tailor-made for your model.

One small correction to your recollection of their connectivity result: Böhm et al. report that **both bursting and regular-firing cells have within-class recurrent excitatory connections**, while cross-class excitatory connections were asymmetric—regular-firing → bursting connections were observed, but not the reverse. Inhibitory connectivity was also biased, with regular-firing cells receiving more inhibitory connections. The Journal of ...

So SUB is not simply a feedforward relay.

It has a potentially interesting architecture:

$$CA1 \to \begin{cases} SUB_{burst} \\ SUB_{regular} \end{cases}$$

with different intrinsic response properties and different local circuit roles.

That suggests a computational interpretation

I would consider modeling the two SUB populations as serving two distinct functions:

Bursting SUB cells = replay-gated output channel

They respond preferentially to the strong, synchronized CA1 activity associated with replay/SWRs. Their bursts are especially effective at driving downstream targets because several spikes arrive closely together. This gives you a natural gating mechanism:

$$\text{ordinary CA1 activity} \to \text{weak SUB output}$$

but

$$\text{compressed/coherent CA1 replay} \to \text{SUB burst} \to \text{strong downstream transmi}$$

That is useful because it means **not every CA1 pattern is written downstream**. Replay events receive privileged access.

That may be the missing computational function you're looking for.

Regular-firing SUB neurons could serve a different purpose

Rather than treating them as another memory store, I would initially use them to stabilize or contextualize the bursting output.

Because they have recurrent connectivity and interact with interneurons, you could use the regular-spiking population as a relatively persistent/local network state:

$$SUB_{RF} \leftrightarrow SUB_{RF}$$

that determines whether a replay-driven burst gets propagated.

Then the network becomes something like:

$$CA1_{\text{replay}} \rightarrow SUB_{burst} \rightarrow EC$$

with

$$SUB_{regular} + I_{SUB}$$

controlling the gain/selection of that transmission.

You don't need to claim that this is exactly what the biological circuit does. For an ICLR paper, you could explicitly say you're extracting a **computational motif** from the anatomy.

And there is experimental motivation for treating bursting as an especially effective communication mode: bursts increase the reliability of synaptic transmission and are generally considered an efficient mechanism for overcoming unreliable individual synapses. The subiculum literature specifically discusses bursting neurons as an output population suited for reliable information transfer. OUP Acade...

The key thing I would *not* do is have SUB store another copy of every CA3 trace

That just moves your original capacity problem downstream.

Instead, let SUB perform **selection/gating**, while the actual long-term representation is created downstream.

That gives:

$$\boxed{CA3 = temporary episodic storage}$$

$$\boxed{CA1 = replay readout}$$

$$\boxed{SUB = replay/event gate}$$

$$\boxed{EC/Cortex = long-term learning}$$

That's a very clean architecture.

So what should happen when replay reaches SUB?

I think the simplest useful mechanism is a **nonlinear transformation from sequence activity into a consolidation signal**.

Suppose CA1 replay produces sequential spikes:

$$x_1, x_2, \dots, x_N.$$

CA1→SUB connections converge onto bursting SUB neurons.

A bursting neuron fires only when the incoming CA1 population activity is sufficiently synchronous/coherent:

$$B_j = H\left(\sum_i w_{ij} x_i - \theta_B\right).$$

Once threshold is crossed, instead of one spike it emits a burst:

$$B_j \to \{t, t+\delta, t+2\delta, \dots\}.$$

Then SUB converts:

$$\text{one compressed CA1 replay}$$

into:

$$\text{strong downstream teaching event.}$$

That is already a useful computation.

You could think of SUB bursting as an **importance amplifier**.

A noisy/non-coherent CA1 event might produce:

$$1 \text{ spike}$$

and do very little downstream.

A valid replay might produce:

$$4-5 \text{ spike burst}$$

and strongly modify EC/cortical synapses.

This has a very nice continual-learning interpretation:

> **Replay quality determines consolidation strength.**

And here Pandey's anti-Hebbian CA1→SUB plasticity could become interesting

You don't necessarily need it to preserve the replay sequence.

Instead it could help make the SUB gate **selective**.

Imagine that a CA1 pattern repeatedly drives a particular SUB bursting neuron.

Under ordinary Hebbian learning:

$$CA1_i \to SUB_j$$

would get progressively stronger every replay.

Eventually that SUB neuron might respond too easily to many related CA1 patterns.

That creates overlap.

With Pandey-like depression-biased anti-Hebbian plasticity, frequently causal inputs can instead weaken.

Computationally this could prevent one SUB neuron from becoming permanently dominated by the most frequently replayed CA1 pattern.

In other words, anti-hSTDP at CA1→SUB could function as a **novelty/competition mechanism**, rather than as a memory-storage mechanism.

That's speculative, but it gives you an experimentally testable role:

$$\text{Does anti-hSTDP increase diversity of SUB recruitment across memories?}$$

You could measure:

$$P(SUB_j|M_i)$$

for different memories M_i.

With no plasticity or Hebbian plasticity, perhaps the same bursting cells get recruited repeatedly:

$$M_1, M_2, M_3 \to B_1, B_2, B_3.$$

With anti-Hebbian plasticity, perhaps subsequent traces recruit different neurons:

$$M_1 \to B_{1-10}$$

$$M_2 \to B_{11-20}$$

$$M_3 \to B_{21-30}.$$

Now you have **pattern separation downstream of CA1** without requiring CA1 itself to perform it.

That would be fascinating if it happens naturally in your simulation.

Then you hit the much more important question: what does EC do?

This is where I think you can abstract aggressively.

You don't need:

CA1 → SUB → EC5 → EC2 → DG → CA3 → cortex

with every known connection.

For the ICLR problem, I'd use perhaps:

$$CA3 \to CA1 \to SUB \to LTM$$

where LTM is an abstract "cortical memory" network.

You could call the intermediate downstream population EC if you want biological flavor, but computationally its role should be clear.

The crucial experiment is:

> **Can replay train a long-term representation sufficiently well that the corresponding CA3 trace can subsequently be removed without losing retrieval?**

That directly tests your cache hypothesis.

This gives you a very clean experimental protocol

Suppose you learn memory M_1.

Initially:

$$M_1 \rightarrow CA3_1.$$

Then replay:

$$CA3_1 \rightarrow CA1 \rightarrow SUB \rightarrow LTM.$$

After n successful replay events, you deliberately erase/compress the CA3 representation:

$$CA3_1 \rightarrow \varnothing.$$

Now give the network a cue to M_1.

If the memory can be retrieved from the downstream store:

$$Cue \rightarrow LTM_1 \rightarrow \hat{M}_1,$$

you have demonstrated genuine **memory transfer**.

Then CA3 can reuse those synapses:

$$CA3_1 \rightarrow CA3_{\text{available}}.$$

Now learn $M_8, M_9, \ldots$.

That turns your current capacity result into something much more interesting.

Your existing network has a capacity of approximately seven traces even after replay-driven compression. ICLR_2027_Submission (1)

But suppose adding consolidation allows:

$$7 \rightarrow 20 \rightarrow 50 \rightarrow \ldots$$

because old CA3 representations can be discarded once transferred.

That is an extremely direct answer to catastrophic forgetting.

The computer-cache analogy becomes surprisingly precise

You could formalize CA3 capacity:

$$C_{CA3} = K$$

memory traces.

When the cache is full:

1. identify a consolidated trace,
2. evict it,
3. reuse its synapses for the next memory.

So:

$$M_1, M_2, \dots, M_K$$

occupy CA3.

Replay transfers:

$$M_1 \to LTM.$$

Then:

$$\text{evict } M_1.$$

Now:

$$M_2, \dots, M_K, M_{K+1}$$

can be stored.

This immediately opens up something very familiar from computer science:

A replacement policy.

Which CA3 memory should be consolidated/evicted?

You could implement:

- oldest first,
- most replayed first,
- strongest downstream transfer first,
- least recently accessed,
- lowest novelty,
- highest cortical reconstruction accuracy.

For ICLR, **this is probably much more compelling than reproducing the complete hippocampal anatomy**.

And now "memory enrichment" gets a simpler interpretation

Previously we were struggling with what exactly enrichment means.

In this architecture, enrichment occurs **after transfer**, not necessarily inside CA3.

CA3 initially stores:

$$M_i.$$

The downstream network has previous knowledge:

$$K.$$

Replay causes:

$$K' = f(K, M_i).$$

This means the new trace changes the distributed long-term representation.

For example:

$$M_1 = [A, B, C, D]$$

$$M_2 = [A, B, C, E]$$

$$M_3 = [A, B, C, F].$$

CA3 treats them as separate episodic sequences.

But the downstream network learns:

$$[A, B, C]$$

as common structure, while maintaining:

$$D, E, F$$

as trace-specific information.

That gives you **enrichment/generalization** without requiring SUB to perform semantic learning.

SUB simply controls which replay events get into the downstream learning system.

There is also a real neuroscience basis for making SUB the gateway

The general SWR literature treats hippocampal sharp-wave/ripple output as a mechanism by which compressed hippocampal representations influence distributed cortical circuits during consolidation. PubMed +1

Böhm et al. make the SUB link much more specific: bursting subicular cells are preferentially recruited during ripples and therefore appear particularly suited to transmit hippocampal ripple information downstream. PubMed +1

There is also evidence that the subiculum is not just a passive conduit. More recent work shows that the subiculum itself can generate ripple-like activity under some conditions, emphasizing that it is a functional network rather than merely an axonal relay.
PubMed Centr...

So a simplified computational interpretation such as

$$CA1\ replay \rightarrow SUB\ burst\ selection \rightarrow downstream\ consolidation$$

is not unreasonable at all.

What I would build next

I would actually **stop at SUB initially**, before building EC.

Your immediate next simulation could be:

$$CA3 \rightarrow CA1 \rightarrow SUB_B + SUB_R + I_{SUB}.$$

Use perhaps:

- bursting and regular-spiking AdEx neurons,
- preferential recruitment of bursting cells by strong CA1 replay,
- local SUB connectivity loosely inspired by Böhm et al.,
- your Pandey-derived anti-hSTDP at CA1→SUB,
- a simple output/readout population receiving SUB bursts.

Then ask only three questions:

1. **Does a valid CA1 replay reliably generate a downstream SUB burst pattern?**
2. **Does that SUB pattern distinguish different memory traces?**
3. **Does CA1→SUB plasticity improve or degrade that distinction across sequential memories?**

The really useful new metric would be **trace separability**:

$$S = \frac{d(\mu_i, \mu_j)}{\sigma_i + \sigma_j}$$

or simply population-vector cosine similarity / Jaccard overlap between SUB responses.

You may discover something very interesting:

$$\text{CA3 traces overlap substantially}$$

but

$$\text{SUB bursting ensembles overlap less.}$$

If so, SUB itself starts providing the feature you were originally looking for: a **replay-triggered sparse code suitable for writing to long-term memory**.

If not, that's also informative—SUB can then remain a pure transmission gate, and you put pattern separation into your abstract cortical/LTM network.

My current working hypothesis would therefore be

$$\boxed{CA3 = finite - capacityepisodiccache}$$

$$\downarrow \text{ replay}$$

$$\boxed{CA1 = temporalreadout}$$

$$\downarrow$$

$$\boxed{SUB = nonlinearreplaygate/amplifier}$$

$$\downarrow$$

$$\boxed{EC/Cortex = durabledistributedstore}$$

and once

$$\text{retrieval fidelity}_{LTM} > \theta,$$

you allow:

$$\boxed{\text{CA3 trace eviction + synaptic reuse}}$$

That last operation is the experiment I think could transform your current ICLR story. Instead of merely showing that replay **delays** interference, you'd be testing whether replay enables a **two-timescale memory architecture that escapes the intrinsic capacity limit of the recurrent hippocampal store**.

And that is immediately recognizable to an ML audience as a continual-learning architecture, even if the SUB/EC implementation is intentionally simplified rather than anatomically exhaustive.

I am curious to know if there are some insight into what activity pattern emerge in the SUB during exploration. Are the RS / BS neurons spiking? Are they spiking on Theta (like in CA3 / CA1)? What about the inhibitory neurons? What about plasticity during exploration?

Yes. There is enough in the in-vivo literature to constrain a useful **exploration-state SUB model**, although it is much less cleanly characterized than CA1. The main picture I get is that during locomotion/exploration, **both regular-spiking (RS) and bursting (BS) subicular principal cells are active, subiculum is strongly theta-modulated, and at least a subset of SUB neurons shows place-related firing and theta phase precession.** The inhibitory population also participates in theta, but the detailed cell-type-specific rules are less complete than in CA1. JN The Journal of ... +2

The part that may be especially useful for your model is that SUB does **not** appear to merely copy CA1 place-cell activity. Its spatial code is more distributed and heterogeneous.

Principal-cell activity during exploration

Kim, Ganguli & Frank (2012) is probably the first paper I would look at closely for your question. They recorded CA1 and subiculum during spatial behavior and found that subicular neurons carry spatial information, but typically with **broader and more distributed firing fields than CA1**. Importantly, subicular neurons nevertheless exhibited clear **theta phase precession**. So SUB output retains a sequential temporal code embedded in theta even though its spatial representation is less sparse/localized than CA1. JN The Journal of ... +1

Conceptually, during exploration you can therefore reasonably have:

$$CA1_{\text{place sequence}} \rightarrow SUB_{\text{spatial activity}}$$

with both populations modulated by roughly the same theta rhythm:

$$\theta \approx 6 - 10 \text{ Hz.}$$

But the SUB representation does not have to look like another narrow CA1 place-field sequence.

Instead, something like this is more defensible:

$$\text{CA1: relatively selective place fields}$$

$$\downarrow$$

$$\text{SUB: broader / overlapping spatial fields + theta phase code.}$$

That difference could actually become useful for your memory-transfer model.

What about bursting versus regular-spiking cells?

Simonnet & Brecht (2019) is particularly valuable because they recorded identified subicular principal cells in freely moving rats and explicitly examined **burst firing during spatial behavior**. JN The Journal of ... +1

One important finding is that bursting is **not restricted to offline replay**.

Subicular principal neurons burst during active exploration as well.

But burst propensity varies considerably between cells. Simonnet and Brecht found that neurons with relatively **sparse bursting** tended to have better spatial coding, while heavily bursting neurons generally had poorer spatial specificity. In other words, bursting isn't simply an "on/off replay mode"; it is part of normal spatial coding during behavior. JN The Journal of ...

That makes me revise one aspect of our previous discussion.

I would **not** model:

$$BS = \text{replay-only cells.}$$

Instead:

$$BS = \text{principal cells capable of nonlinear burst output}$$

that participate during both:

$$\text{exploration}$$

and

$$\text{SWR/replay.}$$

Their probability of bursting would depend on their input and membrane state.

That is much more biologically reasonable.

A useful model distinction

You could therefore let RS and BS cells receive essentially the same CA1 spatial input, but give them different intrinsic dynamics:

RS neuron

$$I_{CA1} \to \text{single spikes}$$

BS neuron

$$I_{CA1} \to \begin{cases} \text{single spike}, & I < I_B \\ \text{burst}, & I > I_B. \end{cases}$$

During normal exploration, CA1 input varies with place field and theta phase, so a BS neuron occasionally bursts.

During compressed replay, CA1 population activity is much more synchronous:

$$I_{CA1}^{replay} \gg I_{CA1}^{exploration},$$

making burst generation much more probable.

That gives you a **single neuronal mechanism** that works in both states.

You wouldn't need to artificially tell the model:

> "Now we are in replay, switch the BS neurons on."

SUB theta is real and quite prominent

The subiculum has strong theta activity during locomotion. Individual SUB principal cells can be theta modulated, and Kim et al. found phase precession within subicular spatial fields much like CA1. JN The Journal of ...

There is also evidence that subicular activity carries more than location. A later study found that subicular neurons can jointly represent environmental context, action/choice and spatial variables, with theta potentially coordinating those different variables. PubMed Centr...

That suggests that the SUB representation may gradually transform:

$$\text{pure spatial sequence}$$

into something closer to

$$\text{location} + \text{context} + \text{behavioral state}.$$

For your first implementation, though, I would **not add those extra variables yet**.

Simply inheriting the theta-modulated spatial drive from CA1 would already be well supported.

What do the interneurons do?

This part is less neatly characterized, but there is useful in-vivo evidence.

Czurkó et al. recorded interneurons across the hippocampal formation in freely moving rats, including subicular regions, and found that interneurons can be strongly **theta phase locked**. Importantly, different interneurons preferred different theta phases rather than behaving as one homogeneous inhibitory population. JN The Journal of ... +1

So during exploration you can reasonably model:

$$I_{SUB}(t) \sim I_0 + A_\theta \cos(2\pi f_\theta t + \phi_I).$$

In other words, SUB inhibition should probably not just be stationary Poisson inhibition.

It can participate in the theta rhythm and constrain when principal cells fire.

That potentially gives you a simple mechanism for producing SUB theta phase selectivity:

$$CA1_\theta + I_{SUB,\theta} \rightarrow \text{phase-restricted SUB spikes}.$$

This is analogous to what you've already been doing with excitatory/inhibitory interactions in CA3, but I would initially make the SUB inhibitory circuit much simpler.

I wouldn't model interneuron subtypes yet

There are PV, somatostatin and other interneuron classes in subiculum, and their targeting is clearly not equivalent. But I don't think reproducing all of that buys you much for ICLR unless it changes your memory-interference result. Inhibitory plasticity in SUB is also cell-type dependent; recent experimental work has found different forms of perisomatic inhibitory plasticity onto different subicular pyramidal-cell classes. Fronti...

For your purposes, one fast-spiking inhibitory population may be enough:

$$SUB_{PC} \rightarrow SUB_I$$

$$SUB_I \rightarrow SUB_{PC}.$$

Use it to:

- stabilize firing,
- generate competition,
- help maintain sparse output,
- synchronize/gate activity with theta.

That's probably sufficient initially.

Plasticity during exploration is where things become particularly interesting

There definitely **is plasticity at CA1→SUB synapses**. It is not simply a fixed relay.

And the intriguing part is that the plasticity rules appear to differ depending on the intrinsic firing phenotype of the postsynaptic SUB neuron.

Wozny et al. showed that CA1→SUB synapses onto bursting and regular-firing cells can exhibit **different forms of long-term potentiation**, with different dependence on NMDA receptors. They interpreted BS and RS cells as functionally distinct output channels with distinct synaptic-plasticity mechanisms. PubMed Centr...

That result is extremely relevant to you.

It means you don't necessarily need:

$$CA1 \rightarrow SUB_{RS}$$

and

$$CA1 \rightarrow SUB_{BS}$$

to follow identical learning rules.

Pandey & Sikdar then showed the depression-biased non-Hebbian STDP rule we've discussed at CA1→SUB synapses. Their results also indicated contributions from muscarinic acetylcholine receptors and voltage-gated calcium channels to bidirectional plasticity. PubMed Centr...

So biologically, CA1→SUB plasticity is probably better thought of as:

$$\Delta W = f(\Delta t, \text{cell type}, \text{activity state}, ACh, Ca^{2+}, \ldots)$$

rather than one universal STDP rule.

That actually gives you considerable freedom computationally.

Exploration versus replay could naturally have different effective plasticity

This is where the Dembitskaya paper you uploaded becomes relevant again.

Their major message is that the effective STDP rule depends heavily on firing pattern, rate and bursting; naturalistic irregular or bursty activity can produce very different plasticity from regularly timed spike pairs. Dembitskaya et al 2026 - Irregu...

They also explicitly note that activity bursts can dominate plasticity even when they represent only a subset of all spikes. Dembitskaya et al 2026 - Irregu...

That gives you a nice distinction:

Exploration

Relatively theta-organized sequential activity:

$$CA1_A \to CA1_B \to CA1_C \to ...$$

with moderate firing and phase precession.

CA1→SUB synapses learn the **online spatial representation**.

Replay

The same sequence becomes highly compressed:

$$A \to B \to C \to ...$$

with much shorter ISIs, more synchrony and more bursting.

Consequently, even with the same underlying biological plasticity machinery,

$$P(\Delta W|\text{replay}) \neq P(\Delta W|\text{exploration}).$$

You don't necessarily have to implement two arbitrary state switches.

That is a useful idea.

For your model, I would implement exploration in SUB surprisingly simply

You could extend your existing online phase like this:

$$EC/DG \to CA3 \to CA1 \to SUB.$$

For every environment:

CA1

Continue essentially what you already have:

$$r_i^{CA1}(x,t) = \text{place modulation} \times \theta(t)$$

or let CA3 drive it if you're now actually modeling the SC pathway.

SUB RS cells

Give them broader convergence from CA1:

$$SUB_j^{RS} \leftarrow \{CA1_{i_1}, CA1_{i_2}, ..., CA1_{i_k}\}.$$

Therefore individual SUB cells naturally acquire broader spatial fields.

SUB BS cells

Same general CA1 input, but with intrinsic bursting dynamics:

$$V > \theta_B \rightarrow 2\text{–}5 \text{ spikes at short ISI.}$$

Don't force them to burst on every activation.

SUB interneurons

Excited by RS/BS principal cells and provide feedback inhibition:

$$RS, BS \rightarrow I \rightarrow RS, BS.$$

Add theta modulation either through their drive or inherited CA1 input.

The really interesting thing to measure would be how representation changes CA1 → SUB

You are thinking mainly about memory transfer, but this gives you a useful intermediate analysis.

For each environment m, compare:

$$R_{CA3}^m, \quad R_{CA1}^m, \quad R_{SUB}^m.$$

Measure:

Population overlap

$$O_{ij}.$$

Spatial information

$$I_{\text{spatial}}.$$

Sparsity

$$S.$$

Theta phase locking

$$R_\theta.$$

Burst fraction

$$B = \frac{N_{\text{spikes participating in bursts}}}{N_{\text{spikes}}}.$$

You might find something like:

$$CA1 : \text{high spatial precision / sparse}$$

while

$$SUB : \text{broader representation / greater redundancy.}$$

That would actually make functional sense for memory transfer.

CA1 could provide a precise episodic sequence, while SUB converts it into a more **redundant and robust output code** suitable for broadcast to EC/cortical targets.

This changes how I think SUB may fit your cache hypothesis

Rather than:

$$CA3 \to CA1 \to SUB \to \text{same exact sequence}$$

I would now consider:

$$\boxed{CA3 = compressed episodic sequence store}$$

↓

$$\boxed{CA1 = high-fidelity sequential readout}$$

↓

$$\boxed{SUB = redundant/distributed hippocampal output representation}$$

↓

$$\boxed{EC/Cortex = long - term representation}$$

That makes the broader fields in SUB potentially a **feature rather than noise**.

A sparse CA1 representation might say:

> location 127 cm in environment A.

A distributed SUB population might convey something closer to:

> this broader portion/context/state of environment A is active.

That downstream representation may be easier to associate with multiple cortical features.

This is where your "memory enrichment" idea may eventually enter.

One thing I would test before adding EC

I think there is an unusually clean experiment available now.

Train one environment and record simultaneously:

$$CA3 \to CA1 \to SUB.$$

Compare **online traversal** and **offline replay** for exactly the same trace.

For each SUB neuron ask:

$$r_{online}, \quad r_{replay}, \quad B_{online}, \quad B_{replay}, \quad \phi_\theta.$$

Then divide the cells into RS and BS.

The prediction I would test is:

During exploration

$$RS + BS$$

both carry the spatial sequence, theta organized.

During replay

become disproportionately recruited because compressed CA1 input is especially effective at driving their intrinsic bursting.

That prediction is directly aligned with the Böhm ripple result and with Simonnet's observation that bursting is already present during spatial behavior. JN The Journal of ... +1

If your simulation reproduces that **without explicitly coding an online/offline switch**, you would have something quite nice:

$$\boxed{\text{same CA1}\!\to\!\text{SUB architecture} + \text{different temporal statistics} \to \text{state-depende}}$$

And that gives SUB a meaningful computational role in your ICLR model: during exploration it participates in ongoing spatial output, but during replay its bursting population becomes a **high-gain channel for exporting the selected memory trace out of the hippocampus**.

I think this is considerably more promising than treating SUB as merely another sequence network.

Another idea that I have is that the SUB Bursting cells, especially the ones that do not display place field tuning behavior, are some kind of a context backbone (like the earlier idea you proposed, that the pattern is not just the distinct new pattern, but rather a generalization pattern S - specific pattern Ps, so the new pattern is S - Ps). I am curious to see if there is any work that show a remapping of the SUB non-place tuned bursting neurons in new environments / new context

Yes — and I think there is **real support for a version of your “context backbone” idea**, although I would phrase it more carefully than “non-place bursting neurons encode context.” The literature does not yet establish that specific cell class/function mapping. What it does show is that SUB population codes are unusually **stable, distributed, and structured across environments**, even when individual neurons remap.

A particularly important old result is **Sharp (1997)**. Unlike CA1 place cells, which tend to remap strongly between sufficiently different environments, subicular cells showed surprisingly similar spatial firing patterns across two geometrically and visually distinct environments. Sharp interpreted this as evidence that SUB may carry a more abstract spatial representation rather than a completely environment-specific map. PubMed +1

That is already quite close to your proposed decomposition:

$$M_i = S + P_i$$

where

- S is some shared/contextual or structural backbone,
- P_i is the environment-specific component.

I would use a **plus** rather than $S - P_i$ for the conceptual decomposition unless you specifically mean a residual representation.

The intriguing possibility is:

$$CA3/CA1 \approx P_i + S$$

but SUB may place relatively more weight on:

$$S$$

than CA1 does.

The bursting result makes this more interesting

Simonnet & Brecht's 2019 recordings are relevant because they found two in-vivo firing regimes among subicular principal neurons. Roughly 80% were sparsely bursting and ~20% were dominantly bursting. The dominantly bursting neurons had **substantially weaker spatial modulation**, whereas sparsely bursting neurons carried more spatial information; all identified boundary cells were in the sparsely bursting population. PubMed Centra... +1

That is compatible with — although it does **not prove** — your idea:

$$\text{dominantly bursting SUB neurons} \neq \text{precise place representation.}$$

The obvious question then becomes:

> If these strongly bursting neurons are firing robustly but are poorly tuned to location, what variable *are* they representing?

Context, environmental identity, behavioral state, memory state, or a low-dimensional structural code are all reasonable candidates.

Simonnet & Brecht did not test your specific hypothesis because their experiment wasn't a systematic multi-environment remapping experiment.

But there is a brand-new 2026 result that I think you should look at

A May 2026 preprint by **Abramson et al.**, *Spatial remapping in the subicular complex and entorhinal cortex follows low-dimensional geometric principles*, is unusually relevant to what you're proposing.

They recorded large populations in SUB/entorhinal circuitry while mice navigated **many distinct environments with widely varying geometries**. At the individual-neuron level, firing patterns looked heterogeneous and individual cells could remap substantially. But at the **population level**, the new-environment representation was not arbitrary. Representations between rooms could often be related by a relatively simple low-dimensional affine transformation — rotations, scaling, shear, reflections, translations. bioR... +1

Their interpretation is essentially that the hippocampal formation retains an underlying **coordinate template** that gets adapted to individual environmental geometry rather than rebuilding an entirely unrelated representation each time. Sci... +1

That strikes me as extremely close to your $S + P_i$ idea.

Their result can almost be written as:

$$R_i = T_i(S),$$

where

- S = shared internal representation,
- T_i = environment-specific transformation.

Or, in your formulation:

$$R_i = S + P_i.$$

The details differ, but computationally the idea is remarkably similar: **new environments need not be represented by completely independent codes**.

Sharp's earlier work goes even further in this direction

Sharp followed the 1997 study with a 2006 paper whose title is almost provocative in this context:

"Subicular place cells generate the same 'map' for different environments: Comparison with hippocampal cells."

The central observation was again that SUB representations were much less prone than hippocampal place cells to globally reorganize between environments. E ScienceDir... +1

Sharp's proposed division of labor was roughly:

$$\text{hippocampus} \rightarrow \text{environment-specific representation}$$

versus

$$\text{subiculum/entorhinal system} \rightarrow \text{more generalized spatial framework.}$$

That's extraordinarily relevant to the computational architecture you're considering.

It suggests that the information exported from CA1 may undergo a transformation something like:

$$\boxed{\text{specific episode} \rightarrow \text{shared structural representation} + \text{episode-specific deviatio}}$$

before being propagated farther downstream.

There's also modern evidence that SUB represents variables beyond location

This is another piece that strengthens your idea.

Lee et al. found that SUB neurons can show **mixed selectivity** for task-related variables rather than simply representing position. SUB population activity represented combinations of spatial location, task phase, action and context, and these representations were more schematic than the punctate CA1 place code. PubMed Centra... +1

Their earlier analysis described subicular firing patterns as **more schematic and less sensitive to exact spatial location** than CA1 representations. PubMed Centr...

That wording — *schematic* — is extremely interesting for your model.

It pushes toward:

$$CA1 : \text{episode/location-specific}$$

$$SUB : \text{task/context/structural representation.}$$

So I wouldn't necessarily call the non-place population "noise."

It may be carrying dimensions that aren't visible when firing is regressed only against x, y position.

What I haven't found is the decisive experiment you are asking for

I don't see a strong published experiment yet that does all three of these simultaneously:

1. physiologically identifies **dominantly bursting SUB cells**,
2. records the same cells across multiple contexts/environments,
3. asks whether the **non-place-tuned bursting population remaps according to environmental identity**.

That would be the clean test of your exact hypothesis.

Simonnet/Brecht give you the **burst phenotype versus spatial tuning**.

Sharp gives you **cross-environment SUB stability/generalization**.

Abramson et al. give you **structured population remapping across many environments**.

But none, as far as I can find, joins those pieces into:

$$\boxed{\text{dominantly bursting non-place SUB cells} = \text{context backbone}}$$

So that remains a hypothesis — but a surprisingly plausible one.

There are actually two different versions of your hypothesis

And I think distinguishing them could help your model.

Hypothesis A: stable backbone

A subset of SUB neurons maintains approximately the same activity relationship across environments:

$$B(E_1) \approx B(E_2) \approx B(E_3).$$

These cells encode something common to all contexts.

Then the environment-specific population adds:

$$P_1,\ P_2,\ P_3.$$

So:

$$R(E_i) = B + P_i.$$

This is close to what I understand you initially proposed.

Hypothesis B: context code

The same non-spatial SUB neurons change their firing rates/patterns between environments:

$$B(E_1) \neq B(E_2) \neq B(E_3),$$

but without becoming place tuned.

Then the population vector identifies **which environment** the animal is in even though it does not specify position.

That's closer to a pure contextual code:

$$C_i = \text{context identity}.$$

Then:

$$R(E_i, x) = C_i + P_i(x).$$

For your memory-interference problem, **B may actually be more useful than A**.

Why context coding would help interference

Suppose your CA3 memory currently looks like:

$$M_i = \{PC_{i1}, PC_{i2}, ..., PC_{in}\}.$$

Different environments are just random reorderings of the same PC population, which is why recurrent synapses eventually collide and saturate. Your current manuscript explicitly creates each new environment by independently randomizing place-cell ordering.

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Now introduce a contextual latent variable:

$$C_i.$$

Your memory becomes:

$$M_i = (C_i, P_i).$$

Then downstream retrieval doesn't require the system to discriminate memories solely from overlapping place sequences.

It can retrieve:

$$C_i \to P_i.$$

That is effectively **conditional memory**.

In machine-learning terms:

$$p(P|C).$$

And that is dramatically easier than storing all P_i in one undifferentiated weight matrix.

But I think the Abramson result suggests an even better formulation

Rather than giving each environment an arbitrary one-hot context code:

$$C_1 = (1, 0, 0, \ldots)$$

$$C_2 = (0, 1, 0, \ldots),$$

you could allow SUB to learn a **shared latent space**.

For example:

$$S = \text{common environmental structure.}$$

Individual memories are then transformations:

$$M_i = T_i(S) + P_i.$$

This is much closer to representation learning.

Imagine your environments are different linear tracks.

Instead of learning:

$$E_1 = [PC_{15}, PC_{72}, PC_{811}, ...]$$

$$E_2 = [PC_{741}, PC_{122}, PC_4, ...]$$

as completely unrelated objects, SUB might extract:

$$S = \text{“linear trajectory”}.$$

Then each experience contributes an embedding:

$$z_i.$$

So:

$$E_i = f(S, z_i).$$

That's **exactly** the kind of factorized representation that tends to reduce interference in ML.

This might also explain why highly bursting SUB cells are weakly spatial

Suppose spatially tuned SUB neurons carry:

$$P(x).$$

Then there is no reason your contextual population should also encode x.

In fact, you would expect:

$$I(B; x) \approx 0$$

but:

$$I(B; C) > 0.$$

Where:

- B = bursting-population activity,
- x = position,
- C = context/environment.

That gives you an incredibly clean prediction.

For a neuron:

$$I_{\text{spatial}} \approx 0$$

does **not** imply:

$$I_{\text{information}} \approx 0.$$

It may instead encode:

$$\text{context}.$$

This is exactly why analyzing only place-field tuning can miss important hippocampal representations.

There is another intriguing fact from Simonnet & Brecht

They found that **dominantly bursting cells comprise only about 20% of the SUB principal population**, while sparsely bursting cells are ~80%. PubMed Centr...

Imagine that computationally:

$$80\% \to \text{specific spatial/output information}$$

$$20\% \to \text{latent contextual/structural representation.}$$

You don't need many context cells.

A relatively small population could encode enormous numbers of contexts through combinatorial codes.

For N binary-ish context neurons:

$$2^N$$

possible states.

Even 100 context neurons provide an enormous representational space.

That gives you a very cheap computational mechanism for indexing memories.

And suddenly CA3 → CA1 → SUB has a very interesting transformation

You could model:

$$\boxed{CA3}$$

as:

high-dimensional episodic trace

↓

$$\boxed{CA1}$$

faithful temporal readout

↓

$$\boxed{SUB}$$

splits the signal:

$$\begin{cases} P_i(x) & \text{sparse/spatial component} \\ C_i & \text{context/latent component} \end{cases}$$

↓

$$\boxed{EC/Cortex}$$

stores:

$$p(P_i|C_i).$$

That is a much more interesting computation than simply forwarding the CA1 sequence.

There's a beautiful experiment hiding here for your ICLR paper

And it doesn't require you to simulate cortex yet.

Take, say, your existing five environments:

$$E_1, \ldots, E_5.$$

Build the SUB population with RS / sparsely bursting / dominantly bursting neurons.

Train through CA1 during exploration and replay.

Then calculate, independently for each SUB cell:

$$I(cell; position)$$

and:

$$I(cell; environment).$$

You would then get four possible cell types:

	Low environment info	High environment info
High spatial info	generic place cell	context-specific place cell
Low spatial info	nonspecific	**context cell**

Your proposed population is that bottom-right quadrant:

$$\boxed{I(position) \approx 0, \qquad I(environment) \gg 0.}$$

Then ask:

> Are these cells disproportionately dominantly bursting?

That gives you an actual quantitative test of your idea.

You could additionally compare:

$$P(BS|context)$$

against

$$P(BS|place).$$

Then test remapping directly

For each neuron j, define an environment response vector:

$$r_j = [\bar{r}_j(E_1), \bar{r}_j(E_2), \ldots, \bar{r}_j(E_N)].$$

A context-selective non-place cell might look like:

$$r_j = [15, 2, 1, 3, 1] \text{ Hz},$$

while having no significant position tuning within E_1.

Another might be:

$$r_k = [1, 2, 14, 2, 1].$$

The population context vector:

$$C(E_i) = [r_1(E_i), ..., r_m(E_i)]$$

could then identify the environment.

Calculate:

$$\text{accuracy}\left(\hat{E}(C)\right).$$

If you can decode the environment from non-place bursting SUB neurons while excluding all spatially tuned cells, **you have discovered exactly the backbone you're hypothesizing.**

There's an even stronger version relevant to enrichment/generalization

Don't make all environments completely unrelated.

Create:

$$E_1, E_2, E_3$$

sharing some structural property, and:

$$E_4, E_5$$

sharing another.

Perhaps:

- linear environments,
- circular environments,
- environments containing common sequence fragments,
- different contexts with overlapping cues.

Then ask whether the latent SUB population clusters:

$$C(E_1) \approx C(E_2) \approx C(E_3)$$

while:

$$C(E_4) \approx C(E_5).$$

Now SUB isn't simply memorizing environment IDs.

It's learning **context classes**.

That gives you:

$$S_k = \text{shared schema}$$

and:

$$P_i = \text{specific episode.}$$

This is much closer to the enrichment question you raised at the start of our discussion.

The very new Abramson study makes this direction particularly timely

Their population-level finding is important because it argues that apparently messy single-cell remapping can contain a **simple latent structure**. Across many differently shaped rooms, SUB/EC population representations could be related using low-dimensional geometric transformations. bR bioR... +1

That's almost an experimental demonstration of the principle:

$$\boxed{\text{high-dimensional neuronal activity} = \text{shared latent structure} + \text{context-depen}}$$

They don't claim that dominantly bursting neurons carry that latent representation. That's your leap.

But it is a **testable and computationally interesting leap**, rather than one contradicted by the literature.

So I would now refine your original idea to something like:

> **Subicular output may decompose replayed hippocampal representations into spatially specific and context/structure-related components. We hypothesize that weakly spatial, strongly bursting subicular neurons contribute to a low-dimensional contextual backbone shared across related experiences, whereas spatially tuned neurons carry episode-specific information.**

That is the version I would be comfortable putting forward as a modeling hypothesis.

And for an ICLR submission, I actually like it much more than building a detailed EC5→cortex anatomical model. It gives you a very ML-relevant concept — **factorized representations** — while still being motivated by some unusually suggestive subiculum physiology and by the newest remapping data.

I think there is one thing that I would like to show that will be really cool for ICLR, and in general for the AI industry. The issue is not the SUB, but the EC output. If SUB can encode context and place trajectory (e.g. the new memory trace within context) then I will need to show that it can help with interference with being able to replay or provide a cued recall for a large number of environments (more than the CA3 limit). EC 2/3 provides input to CA3 / CA1 and I think also to SUB. In addition, EC2/3 provides input to EC5. SUB also provides input to EC5. It would be good

Show more

Yes — this is much closer to the experiment I think could make the paper stand out for ICLR. The key advance would not be "model more hippocampal anatomy," but to demonstrate **memory export followed by context-conditioned prediction outside CA3**.

Your current manuscript establishes that CA3 behaves like a finite-capacity fast store: sequential memories saturate recurrent connectivity, and even replay-driven compression only extends capacity to around seven traces. ICLR_2027_Submission (1) What you are proposing gives replay a qualitatively new role:

$$\boxed{\text{Replay is the mechanism that moves a memory from a limited fast store into a}}$$

I think that is a considerably stronger continual-learning story.

One anatomical correction is worth making before designing it. The canonical superficial EC input is broadly:

$$ECII \rightarrow DG/CA3$$

and

$$ECIII \rightarrow CA1/SUB.$$

CA1 and SUB then project strongly back to **deep EC**, particularly layer V. PubMed Centra... +1 Layer V is not homogeneous, however. Work separating MEC layer Va and Vb shows that deep EC contains different projection populations and substantial circuitry back toward superficial EC. Subicular input can converge onto MEC layer-V neurons that themselves project to superficial layers. PubMed So your proposed loop is reasonable at an abstract level, but I would probably model it as:

$$EC_{sup} \rightarrow HC$$

and

$$SUB \rightarrow EC_{deep}$$

with an intracortical:

$$EC_{deep} \rightarrow EC_{sup}$$

feedback path rather than assuming a simple homogeneous EC2/3→EC5 connection.

And that loop is exactly what makes your prediction idea interesting.

What I think EC5 should learn

Suppose environment m consists of a context C_m and trajectory

$$P_m = (p_1, p_2, \dots, p_T).$$

Your existing hippocampal pathway initially learns the experience:

$$EC_{sup} \to CA3 \to CA1 \to SUB.$$

For the moment, assume SUB produces something like the representation we discussed:

$$z_t^{SUB} = [C_m, p_t].$$

During offline replay, this sequence becomes:

$$[C_m, p_1] \to [C_m, p_2] \to [C_m, p_3] \to \dots$$

and is delivered repeatedly to EC5.

Now the purpose of EC5 is **not to memorize another sequence using the same recurrent mechanism as CA3**.

Instead, EC5 learns the transition function:

$$\boxed{P(p_{t+1} \mid p_t, C_m)}$$

or more generally

$$\boxed{\hat{z}_{t+1} = f_\theta(z_t, C).}$$

That is your “predictor cell” idea.

I think that is exactly the right abstraction.

This would solve a very different problem from CA3

CA3 stores:

$$M_m = (p_1, p_2, \dots, p_T)$$

in its recurrent connectivity.

Every new sequence therefore consumes recurrent synaptic capacity.

EC5 instead learns:

$$(C_m, p_t) \to p_{t+1}.$$

That is a **conditional transition model**.

Now a partial cue:

$$[C_m, p_1]$$

can produce:

$$\hat{p}_2.$$

Then:

$$[C_m, \hat{p}_2] \to \hat{p}_3$$

and so forth.

Therefore:

$$\boxed{\text{cue} \to p_1 \to p_2 \to p_3 \to \dots}$$

can occur even if CA3 no longer contains the original trace.

That would be genuine systems-level memory transfer.

There is also neuroscience precedent for treating EC/hippocampal circuitry predictively

A particularly relevant recent paper is Chen et al. 2024, **"Predictive sequence learning in the hippocampal formation."** They built a predictive autoencoder interpretation of the entorhinal–hippocampal circuit in which sequence learning is formulated explicitly as prediction of future inputs. E ScienceDir... +1

Their architecture is not the same as the one you're proposing, but the computational premise is useful:

$$\text{current representation} \to \text{prediction of future representation.}$$

There are also spiking-network models showing that local Hebbian learning plus homeostasis can create **context-specific prediction of future sequence elements** and autonomous replay. Bouhadjar et al. demonstrated exactly this type of computation in sparse spiking networks: sequence-specific subnetworks learn context-dependent next-state prediction and can later autonomously replay the learned sequence. PubMed Centr...

For ICLR, that literature gives you a very natural bridge:

> Hippocampal replay provides the training data; deep EC learns the predictive model.

That would be a pretty compelling connection between biological memory consolidation and self-supervised sequence learning.

The context component is what could make it resistant to interference

This is where your SUB idea becomes important.

Suppose two environments contain an overlapping trajectory segment:

$$E_A: \quad 1 \to 2 \to 3 \to 4$$

and

$$E_B: \quad 1 \to 2 \to 7 \to 8.$$

If EC5 only sees the current position 2, there is an ambiguity:

$$2 \to 3?$$

or

$$2 \to 7?$$

This is exactly an interference problem.

But SUB supplies context:

$$[C_A, 2] \to 3$$

versus

$$[C_B, 2] \to 7.$$

Thus the transition becomes:

$$\boxed{P(p_{t+1} \mid p_t, C)}$$

rather than:

$$P(p_{t+1} \mid p_t).$$

That is a major computational difference.

It lets multiple conflicting sequences reuse the same representation for p_t without destructive interference.

In ML terms, you're introducing **conditional computation**.

I would deliberately create conflicting sequences

This could make the experiment much stronger than your current randomized-place-field environments.

Instead of simply generating ten completely unrelated environments, construct a controlled interference benchmark.

For example:

$$A: \ 1 \to 2 \to 3 \to 4 \to 5$$

$$B: \ 1 \to 2 \to 6 \to 7 \to 8$$

$$C: \ 1 \to 2 \to 9 \to 10 \to 11$$

$$D: \ 1 \to 2 \to 12 \to 13 \to 14.$$

Without context:

$$1 \to 2 \to ?$$

is increasingly ambiguous.

With context:

$$(C_A, 2) \to 3$$

$$(C_B, 2) \to 6$$

etc.

Now you can quantitatively show:

$$\text{interference}_{no-context} \uparrow$$

with number of memories, while

$$\text{interference}_{context-conditioned} \approx \text{low}.$$

That is immediately understandable to an ML reviewer.

I think EC5 prediction is more interesting than EC5 replay

You mentioned two possibilities:

1. EC5 itself replays the sequence.
2. EC5 simply potentiates/predicts the expected next state after a cue.

I would prioritize **#2**.

If you build another recurrent replay network in EC5, a reviewer can reasonably ask:

> Why doesn't EC5 eventually suffer exactly the same recurrent-weight saturation as CA3?

You've simply relocated your problem.

Prediction allows a different representation.

CA3:

$$\text{sequence stored in recurrent dynamics.}$$

EC5:

$$\text{sequence stored as conditional transitions.}$$

Then retrieval can still *look* like replay if predictions are fed recursively:

$$p_1 \to \hat{p}_2 \to \hat{p}_3 \to \dots$$

but conceptually it is **autoregressive generation**, not CA3-style attractor replay.

That distinction is very useful.

Your EC2/3 input could provide the cue

This also gives a clear function to superficial EC during retrieval.

During original exploration:

$$EC_{sup}(C, p_t) \to HC$$

creates the episodic trace.

After consolidation:

$$EC_{sup}(C, p_t) \to EC5$$

provides the retrieval cue.

EC5 has learned:

$$f(C, p_t) = p_{t+1}.$$

So many simulated days/environments later you present:

$$[C_A, p_1]$$

and ask what EC5 produces.

A successful consolidated trace would generate:

$$p_2, p_3, \ldots, p_T$$

even after the corresponding CA3 trace has been erased.

That is the experiment.

The most convincing result would be CA3 ablation/eviction

I would make this absolutely central.

Stage 1 — Encoding

Environment E_1:

$$EC \to CA3 \to CA1 \to SUB.$$

CA3 can replay it.

Stage 2 — Consolidation

Allow perhaps N replay events:

$$CA3 \to CA1 \to SUB \to EC5.$$

EC5 learns:

$$(C_1, p_t) \to p_{t+1}.$$

Stage 3 — Verify downstream memory

Give EC the cue:

$$(C_1, p_1).$$

Verify EC5 predicts:

$$p_2, p_3, \ldots.$$

Stage 4 — Evict CA3 memory

Remove the CA3 synaptic contribution associated with E_1.

This is important.

You're now explicitly implementing:

$$\boxed{\text{cache eviction.}}$$

Stage 5 — Learn additional memories

Use the freed CA3 capacity for:

$$E_8, E_9, E_{10}, \ldots$$

Stage 6 — Recall an ancient memory

After perhaps 20 or 50 subsequently learned environments, cue:

$$(C_1, p_1).$$

If EC5 still produces:

$$p_2, p_3, \ldots, p_T,$$

then you have demonstrated:

$$\boxed{\text{long-term retention beyond hippocampal capacity.}}$$

That would be a much stronger result than extending CA3 from five to seven memories.

Your figure could be spectacularly simple

Imagine the x-axis:

$$\text{number of sequentially learned environments}$$

and y-axis:

$$\text{recall accuracy.}$$

Three curves:

CA3 only

drops around five.

CA3 + replay compression

drops around seven.

CA3 → predictive EC consolidation

stays high:

$$5, 10, 20, 30, \ldots$$

depending on how far you test it.

That single figure communicates the whole story.

Your current result provides the first two curves already. ICLR_2027_Submission (1)

You would be adding the third.

And you can make the continual-learning comparison exceptionally clean

Define recall after learning N memories:

$$R_i(N)$$

for every memory i.

Then measure average forgetting:

$$F_i = \max_{n<N} R_i(n) - R_i(N).$$

This is essentially the same sort of forgetting measure commonly used in continual learning.

Compare:

$$F_{CA3}$$

$$F_{CA3+compression}$$

$$F_{CA3+consolidation}.$$

Then the neuroscience model suddenly produces a standard ML result:

> Replay-mediated consolidation dramatically reduces catastrophic forgetting.

That framing would make the connection to ICLR much more explicit than simply counting hippocampal replay events.

The predictor cells don't even need to correspond to literal biological "predictor neurons"

I would avoid asserting:

> EC5 contains predictor cells.

You don't need that claim.

Instead say something like:

> **We hypothesize that hippocampal replay can train downstream entorhinal representations to encode context-conditioned transitions between sequential states.**

Then your EC5 neurons are a computational implementation of that hypothesis.

Their activation can represent:

$$\hat{p}_{t+1}.$$

That's entirely acceptable for ICLR.

The anatomy motivates:

$$SUB \to EC_{deep}$$

and deep EC's role in routing hippocampal output back toward cortical and superficial EC circuitry. Deep EC is a major target of CA1/subicular hippocampal output, and deep-layer activity has been associated with successful subsequent memory. Nature

There is also evidence that deep MEC neurons receiving subicular input can project back toward superficial layers, giving you a plausible recurrent hippocampal–entorhinal loop. PubMed

That's enough biological grounding.

One modification could make this particularly powerful: separate “state” from “context”

Suppose SUB provides:

$$S_t = [C, p_t].$$

Don't give EC5 a dedicated neuron for every (C, p) combination.

Instead encode them separately:

$$z_t = z_C + z_{p_t}.$$

Then EC5 learns:

$$(z_C, z_{p_t}) \to z_{p_{t+1}}.$$

Now you've implemented **factorized memory**.

That creates opportunities for transfer/generalization.

Suppose environments A and B share:

$$1 \to 2 \to 3.$$

EC5 doesn't necessarily need to independently relearn that transition.

It can reuse:

$$p_1 \to p_2$$

while the context signal disambiguates the points where trajectories diverge.

You could quantify how many unique synapses are required per additional environment.

If the marginal synaptic cost falls as related experiences accumulate:

$$\frac{dW}{dM} \downarrow,$$

you have moved from merely avoiding forgetting to **knowledge reuse**.

That is where your earlier “enrichment” idea reappears naturally.

There is an even cooler test

Train:

$$A: \quad 1 \to 2 \to 3 \to 4$$

and:

$$B: \quad 1 \to 2 \to 3 \to 5.$$

Then train a new environment:

$$C: \quad 1 \to 2 \to 3 \to 6.$$

Measure how many replay events are required for EC5 to learn C.

If EC5 already knows:

$$1 \to 2 \to 3,$$

only:

$$3 \to 6$$

is genuinely novel.

Therefore consolidation should become faster:

$$T_{learn}(C) < T_{learn}(A).$$

Now replay isn't merely preventing forgetting.

It is producing **forward transfer**.

That is a very attractive AI result.

I think the architecture is becoming quite coherent

I'd now draw your model as:

$$\boxed{EC_{sup}}$$

sensory/spatial state + context

$$\downarrow$$

$$\boxed{CA3}$$

fast, interference-prone episodic cache

$$\downarrow$$

$$\boxed{CA1}$$

high-fidelity hippocampal readout

$$\downarrow$$

$$\boxed{SUB}$$

factorizes/exports:

$$\text{context} + \text{trajectory}$$

$$\downarrow \quad \text{during replay}$$

$$\boxed{EC5}$$

learns:

$$P(next\ state \mid current\ state, context)$$

and during future retrieval:

$$EC_{sup}\ cue \rightarrow EC5\ prediction.$$

Optionally:

$$EC5 \rightarrow EC_{sup}$$

closes the generative loop.

Then:

$$\text{successful consolidation} \rightarrow \text{CA3 eviction.}$$

That is a complete computational memory architecture.

The single experiment I would prioritize

I wouldn't start by modeling all of this simultaneously.

I would take the **SUB output you already envision as given**:

$$[C, p_t].$$

Don't worry yet about proving biologically how SUB computes it.

Feed those representations into a small EC5 predictor.

Train EC5 **only from replay**, never directly from the online training data.

Then test:

> After removing a memory from CA3, can an EC cue recover its trajectory from EC5 after 10, 20, 50 intervening environments?

Run two versions:

$$EC5(p_t)$$

versus

$$EC5(C, p_t).$$

Deliberately construct overlapping/conflicting trajectories.

If you get:

$$\text{without context} \rightarrow \text{catastrophic interference}$$

but

$$\text{with context} \rightarrow \text{stable old-memory prediction,}$$

I think you have the central result.

It directly demonstrates your broader hypothesis:

$$\boxed{\text{The hippocampus need not solve continual learning by indefinitely increasing}}$$

Instead:

$$\boxed{\text{Replay converts temporary episodic traces into context-conditioned predictive}}$$

That is a much bigger ICLR idea than simply finding another plasticity rule that raises CA3 capacity from seven to nine. It connects replay, memory consolidation, predictive representation learning, catastrophic forgetting, and finite memory buffers in one experiment.

You mentioned EC5 (deep) trajectory back to EC 2/3 (Superficial). How would these connection enhance my work?

The deep-EC → superficial-EC pathway could add something quite important to your model, but I would use it for **closed-loop retrieval**, not merely as an anatomical embellishment.

The anatomy supports the existence of this loop. Hippocampal output from CA1/subiculum targets deep EC, and deep-layer neurons—especially layer Vb—project locally to Va and also back toward superficial EC layers II/III. PubMed +1 There are also MEC layer V neurons that receive subicular input and project to superficial layers, providing a direct way for hippocampal output to influence the very EC populations that feed the hippocampus. PubMed Centr...

For your project, that means EC5 does not have to be the terminal store. It can become the **predictive state** that feeds its prediction back into the hippocampal input stream.

So rather than:

$$EC_{2/3} \to CA3 \to CA1 \to SUB \to EC5$$

ending there, you get:

$$\boxed{EC_{2/3} \to CA3 \to CA1 \to SUB \to EC5 \to EC_{2/3}}$$

That closed loop gives you several experimentally useful capabilities.

1. It lets one prediction generate the next prediction

Suppose a consolidated memory is:

$$(C_A, p_1) \to p_2 \to p_3 \to p_4.$$

EC5 has learned:

$$f(C_A, p_t) = p_{t+1}.$$

A cue arriving in superficial EC activates:

$$[C_A, p_1].$$

EC5 predicts:

$$\hat{p}_2.$$

If EC5 feeds that prediction back to EC2/3, then superficial EC now contains:

$$[C_A, \hat{p}_2].$$

That becomes the input for the next prediction:

$$EC5([C_A, \hat{p}_2]) \to \hat{p}_3.$$

And recursively:

$$p_1 \to p_2 \to p_3 \to p_4.$$

So the EC5→EC2/3 pathway turns a **one-step predictor** into an **autoregressive memory generator**.

That is potentially much better than putting recurrent PC→PC connectivity into EC5.

You can have:

$$\boxed{\text{recurrence through layers rather than recurrence within one population}}$$

which is also closer to the anatomical organization.

---

2. You can replay an old trace after deleting it from CA3

This is probably the strongest application for your paper.

After consolidation:

$$CA3(M_1) \to SUB(M_1) \to EC5$$

you deliberately erase M_1 from CA3.

Much later, give EC2/3 only:

$$C_1 + p_1.$$

If the loop produces:

$$EC_{2/3}(p_1) \to EC5(p_2) \to EC_{2/3}(p_2) \to EC5(p_3) \to \dots$$

you have recreated a sequence **without CA3 generating it**.

That would be a very strong demonstration that the memory has genuinely migrated.

You could even call the resulting activity downstream replay, although for the paper I might describe it more precisely as **predictive sequence reconstruction**.

The distinction would then be:

$$\text{CA3 replay} = \text{recurrent attractor dynamics}$$

versus

$$\text{EC replay} = \text{iterative prediction through a deep–superficial loop.}$$

That difference is scientifically interesting in itself.

---

3. It gives you a clean separation between episodic storage and generative retrieval

Your model could now have two fundamentally different memory systems.

CA3

$$M_i = \text{stored recurrent trajectory.}$$

It learns quickly but suffers interference.

Deep/superficial EC loop

$$f_\theta(C, p_t) \to p_{t+1}.$$

It learns more slowly from replay, but retrieval is generative.

That maps beautifully to the cache analogy:

$$\boxed{\text{CA3} = \text{fast cache}}$$

and

$$\boxed{\text{EC} = \text{compiled predictive model}}.$$

Replay is then analogous to **compilation**:

$$\text{explicit episodic trace} \to \text{predictive transition model.}$$

Once compilation succeeds, the original cache entry can be discarded.

That is a much stronger computational story than simply moving the same representation from CA3 into another matrix.

It also helps enormously with your context idea

Consider two conflicting trajectories:

$$A: \quad 1 \to 2 \to 3$$

$$B: \quad 1 \to 2 \to 7.$$

If EC5 sees only:

$$p_2,$$

the prediction is ambiguous.

But if superficial EC maintains context:

$$EC_{sup} = [C, p],$$

then:

$$EC5(C_A, p_2) \to p_3$$

while

$$EC5(C_B, p_2) \to p_7.$$

Now imagine EC5 sends only the predicted state p_{t+1} back upward while the contextual activity in EC2/3 persists:

$$[C_A, p_2] \to p_3 \to [C_A, p_3].$$

This gives you a **persistent context + evolving trajectory** architecture.

Computationally:

$$C_{t+1} = C_t$$

$$p_{t+1} = f(C_t, p_t).$$

I think that would fit your proposed SUB output extremely well.

SUB might initially deliver:

$$[C_i, p_t]$$

during replay.

After consolidation, EC2/3 holds or reinstates C_i, while EC5 recursively predicts the changing p_t.

That separation may be the key to storing many trajectories without interference.

This gives you a particularly nice experimental comparison

I would compare three downstream architectures.

Model A — no consolidation

$$CA3\ only.$$

Expected capacity:

$$\sim 5$$

or ~ 7 with your compression protocol.

Model B — EC5 one-step prediction

$$[C, p_t] \to p_{t+1}.$$

Test one-step prediction accuracy for old memories.

Model C — EC5↔EC2/3 closed loop

$$[C, p_1] \to p_2 \to p_3 \to \dots.$$

Test complete trajectory reconstruction.

Model C lets you ask the much stronger question:

> Can a partial cue reconstruct an entire memory that disappeared from CA3 tens of environments ago?

That is likely the figure I would want in an ICLR paper.

It may also give you error correction

There is another advantage.

Suppose EC5 predicts:

$$\hat{p}_3$$

but somewhat noisily.

When that activity returns to superficial EC, EC2/3 does not necessarily need to reproduce the exact noisy continuous vector. You can impose sparsification / winner-take-all / attractor-like selection:

$$\hat{p}_3 \to p_3^*.$$

Then feed the cleaned state back into EC5.

So:

$$EC5 \to \underbrace{EC_{sup}}_{\text{state cleanup}} \to EC5.$$

This prevents autoregressive error accumulation.

In ML terms, superficial EC becomes something like a **latent-state quantizer**.

That could be incredibly useful because pure autoregressive prediction tends to drift:

$$\epsilon_1 \to \epsilon_2 \to \epsilon_3 \to \dots.$$

A discrete/sparse EC2/3 representation can reset the prediction onto a valid memory state after every step.

You don't even need to claim that this exact operation is established biologically; it is a biologically inspired computational function.

And this opens the door to hippocampal re-entry

There is a second possible use of the loop that I think is worth keeping in reserve.

Superficial EC sends information back into the hippocampus:

$$ECII \to DG/CA3$$

and

$$ECIII \to CA1/SUB.$$

So a generated EC prediction could eventually do:

$$EC5 \to EC_{sup} \to CA3/CA1.$$

That means a **consolidated downstream memory could reactivate the hippocampus**.

This could provide:

Memory reinstatement

A cortical/EC memory generates:

$$p_1 \to p_2 \to p_3,$$

which drives corresponding CA1/CA3 cells.

Relearning

If the CA3 representation was removed, downstream prediction could retrain it:

$$EC \to CA3.$$

Memory enrichment

Suppose an old consolidated representation predicts:

$$p_1 \to p_2 \to p_3.$$

The animal now experiences something new:

$$p_3 \to p_8.$$

The existing EC representation can seed a new CA3 episode containing both old structure and new information.

That gives you:

$$\boxed{\text{old knowledge} + \text{new episode} \to \text{enriched hippocampal trace}}$$

which starts to answer your original enrichment question.

I wouldn't implement this in the first version, but it follows naturally from the architecture.

There is useful anatomical support for the loop, with an important nuance

The classical picture of EC as simply:

$$\text{superficial EC} \to HC$$

and

$$HC \to \text{deep EC} \to cortex$$

is too simple. Modern tracing studies show intrinsic deep-to-superficial pathways. In particular, EC layer Vb neurons innervate Va, II, and III in both medial and lateral EC.

ScienceDir...

There are also layer-V neurons that simultaneously receive subicular and retrosplenial input and project toward superficial MEC, which is precisely the type of circuitry you would want if hippocampal replay is being integrated with broader contextual information before being fed back into the hippocampal input pathway. PubMed Centr...

However, there are meaningful MEC/LEC differences: the Vb→Va pathway appears stronger in LEC than MEC, suggesting that the exact circuitry supporting systems consolidation may differ across EC subdivisions. eL...

For an ICLR model, I would abstract all of that into:

$$EC_D \to EC_S$$

rather than commit heavily to Va/Vb/MEC/LEC details.

One architecture I think is especially elegant

Your eventual model could have four representations:

$$\boxed{C}$$

context

$$\boxed{P_t}$$

current trajectory state

$$\boxed{\hat{P}_{t+1}}$$

prediction

$$\boxed{E_t}$$

prediction error.

Then:

$$EC_S = [C, P_t]$$

$$EC_D = f(C, P_t) = \hat{P}_{t+1}$$

and after observing/reconstructing the next state:

$$E_t = P_{t+1} - \hat{P}_{t+1}.$$

During initial learning, CA3 stores large prediction errors:

$$|E_t| \gg 0.$$

As EC learns the sequence:

$$|E_t| \to 0.$$

Now you get a beautiful **criterion for CA3 eviction**:

$$\boxed{\text{evict memory when EC prediction error falls below } \epsilon.}$$

That's much better than arbitrarily saying “replay five times and delete the CA3 trace.”

The system itself knows when a memory is consolidated.

That gives you:

$$\text{high prediction error} \to \text{keep in CA3}$$

$$\text{low prediction error} \to \text{safe to evict.}$$

This is almost exactly what you would design for a machine-learning memory hierarchy.

And that could lead to the killer ICLR result

Imagine a continual stream of 100 environments.

CA3 capacity:

$$K = 7.$$

Every environment enters CA3.

Replay trains EC.

When:

$$L_{\text{prediction}}(M_i) < \epsilon,$$

M_i gets evicted from CA3.

Then after learning all 100 environments:

CA3 holds only ~7 recent/unconsolidated traces.

But EC can recall:

$$90+$$

old trajectories.

And you compare:

$$\text{CA3 only}$$

vs.

$$\text{CA3} + \text{replay compression}$$

vs.

$$\text{CA3} + \text{replay-driven predictive consolidation.}$$

If the third architecture retains old memories while continually recycling CA3 capacity, you have moved from a neuroscience observation about replay to a **general continual-learning architecture**.

So yes: I think **EC5 → EC2/3 substantially enhances your idea**. It changes EC5 from a passive destination for consolidated information into part of a generative loop:

$$\boxed{\text{cue} \rightarrow \text{prediction} \rightarrow \text{reinstated state} \rightarrow \text{next prediction.}}$$

And perhaps most importantly, that gives you a way to demonstrate **full cued sequence recall beyond the capacity of CA3**, rather than merely showing that some downstream synaptic weights still contain information about an old trace.

Good discussion. Please provide all references mentioned here and also a summary of the last key points / experiments we discussed.

Absolutely. I would preserve the last part of this discussion as the core research direction, because I think it gives you a much cleaner ICLR story than trying to make every intermediate hippocampal structure itself solve interference.

Your current manuscript already establishes the first half of that story: **CA3 has finite capacity under continual learning**, and replay-driven compression delays but does not remove that limit. ICLR_2027_Submission (1) The extension we converged on is essentially a **hierarchical memory architecture in which replay exports traces out of CA3 and trains a context-conditioned predictive model downstream**.

Consolidated working hypothesis

The architecture we ended up with is approximately:

$$\boxed{EC_{sup}} \rightarrow \boxed{CA3} \rightarrow \boxed{CA1} \rightarrow \boxed{SUB} \rightarrow \boxed{EC_{deep}} \rightarrow \boxed{EC_{sup}}$$

with very different computational roles at each stage.

CA3 is the fast episodic cache. It learns individual traces quickly, but because memories share the same recurrent PC–PC matrix, interference eventually becomes unavoidable. That is exactly what your current results demonstrate. ICLR_2027_Submission (1)

CA1 is primarily a downstream readout of CA3 replay rather than another autonomous recurrent memory store.

SUB may transform the CA1 representation into something like:

$$M_i = (C_i, P_i),$$

where C_i is contextual/generalized information and P_i is the specific trajectory or episode component. The evidence does not establish that this is literally what bursting SUB neurons do, but several findings make it a reasonable modeling hypothesis: dominantly bursting cells tend to carry substantially less precise spatial information, SUB representations are more distributed than CA1, and SUB populations can encode context/task variables in addition to position. PubMed +2

Deep EC, rather than simply storing another copy of the sequence, learns a predictor:

$$\boxed{\hat{P}_{t+1} = f(C, P_t)}$$

or probabilistically,

$$P(P_{t+1} \mid P_t, C).$$

The deep→superficial EC pathway then converts a one-step predictor into a recursive generator:

$$[C, P_1] \rightarrow P_2 \rightarrow [C, P_2] \rightarrow P_3 \rightarrow [C, P_3] \ldots$$

Anatomically, this is at least a reasonable abstraction: hippocampal output from CA1/SUB preferentially reaches deep EC, and layer Vb has intrinsic projections back toward layers III and II. Nature +2

That means the old memory can eventually be reconstructed **without CA3 generating its replay**.

The key experiments we converged on

Experiment 1 — Establish the CA3 capacity baseline

You largely already have this.

Compare:

$$CA3 + sSTDP$$

against:

$$CA3 + \text{replay-driven compression.}$$

Your present result is approximately:

$$5 \text{ traces} \rightarrow 7 \text{ traces}$$

before replay fails. ICLR_2027_Submission (1)

That establishes the fundamental problem:

> Improving synaptic efficiency inside CA3 helps, but finite recurrent capacity remains.

Experiment 2 — Replay-mediated export

For a newly learned environment E_i:

$$EC_{sup} \rightarrow CA3 \rightarrow CA1 \rightarrow SUB.$$

During offline replay:

$$CA3 \rightarrow CA1 \rightarrow SUB \rightarrow EC_{deep}.$$

The crucial restriction would be:

ECdeep learns the trace only from replay, not directly from the original exploration input.

That makes replay causally responsible for consolidation.

SUB output can initially be abstracted rather than completely biologically derived. For example:

$$z_t^{SUB} = [C_i, P_t].$$

You can later make the SUB implementation more sophisticated.

Experiment 3 — Context-conditioned next-state prediction

This is probably the most important new experiment.

Train ECdeep to learn:

$$(C_i, P_t) \to P_{t+1}.$$

Then deliberately create trajectories that interfere if context is absent.

For example:

$$A : 1 \to 2 \to 3 \to 4$$

$$B : 1 \to 2 \to 7 \to 8$$

$$C : 1 \to 2 \to 9 \to 10.$$

A context-free predictor sees:

$$2 \to ?$$

and develops interference.

A context-conditioned predictor sees:

$$(C_A, 2) \to 3$$

$$(C_B, 2) \to 7$$

$$(C_C, 2) \to 9.$$

This gives you a beautiful ablation:

$$f(P_t)$$

versus

$$f(C, P_t).$$

If context dramatically increases memory capacity, you have a direct mechanism explaining how a downstream representation avoids the interference experienced by CA3.

Experiment 4 — Close the deep/superficial EC loop

Now let:

$$EC_{deep} \to EC_{sup}$$

feed the predicted state back.

Start with only:

$$[C_i, P_1].$$

Deep EC predicts:

$$\hat{P}_2.$$

Superficial EC reinstates:

$$[C_i, \hat{P}_2].$$

Deep EC predicts:

$$\hat{P}_3$$

and so forth.

The resulting sequence:

$$P_1 \to P_2 \to P_3 \dots$$

looks like replay, but computationally it is very different from CA3 replay.

CA3:

$$\text{recurrent attractor/sequence dynamics.}$$

EC:

$$\text{iterated prediction.}$$

I think that distinction is very attractive for ICLR.

Experiment 5 — CA3 memory eviction

This is probably the most important demonstration.

Once EC can reconstruct memory M_i, deliberately eliminate the CA3 representation.

Then learn more environments using the recycled CA3 capacity.

Much later, cue:

$$[C_i, P_1].$$

If the EC loop still reconstructs:

$$P_2, P_3, \ldots, P_T,$$

you have demonstrated that the memory is no longer dependent on CA3.

That is genuine **memory transfer**, not just another measure of replay preservation.

The conceptual model becomes:

$$\boxed{\text{CA3} = \text{cache}}$$

and

$$\boxed{\text{EC} = \text{durable predictive memory}}.$$

Experiment 6 — Scale beyond the CA3 capacity limit

This should produce the main ICLR figure.

X-axis:

$$\text{number of sequential memories learned}$$

Y-axis:

$$\text{recall accuracy}.$$

Compare three conditions:

$$\text{CA3 only}$$

$$\text{CA3} + \text{compression}$$

$$\text{CA3} + \text{replay-driven predictive consolidation}.$$

Ideally you see something like:

$$5, \quad 7, \quad 20/50/100+$$

rather than trying to eke another one or two memories out of CA3.

That would substantially change the interpretation of your current result. Instead of asking:

> How can CA3 store indefinitely more memories?

the paper asks:

> **Why should CA3 have to?**

Replay provides a mechanism for moving information out of a limited fast store.

Experiment 7 — Prediction error as the eviction criterion

This was one of the more interesting extensions.

Rather than arbitrarily saying that a trace is consolidated after five replay events, calculate:

$$L_i = \sum_t \|P_{t+1} - \hat{P}_{t+1}\|^2.$$

Keep the trace in CA3 while:

$$L_i > \epsilon.$$

Once:

$$L_i < \epsilon,$$

the downstream system can reconstruct it reliably and the trace becomes eligible for eviction.

Thus:

$$\text{high prediction error} \rightarrow \text{retain in CA3}$$

and:

$$\text{low prediction error} \rightarrow \text{evict/recycle CA3 capacity.}$$

This gives you an endogenous memory-management rule remarkably similar to a computer memory hierarchy.

Experiment 8 — Forward transfer / memory enrichment

Once EC has learned shared transitions, new related memories should require less learning.

For example:

$$A : 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$$

$$B : 1 \rightarrow 2 \rightarrow 3 \rightarrow 5$$

$$C: 1 \to 2 \to 3 \to 6.$$

The system already knows:

$$1 \to 2 \to 3.$$

Only:

$$3 \to 6$$

is novel for C.

You can measure:

$$T_{\text{consolidate}}(C) < T_{\text{consolidate}}(A).$$

That moves the result beyond catastrophic-forgetting prevention into **forward transfer and schema reuse**.

This may ultimately provide your operational definition of memory enrichment.

Where SUB fits

I would deliberately avoid making the paper depend on proving exactly how SUB generates context.

There are good reasons to investigate it, but you don't need the entire hypothesis solved before testing downstream consolidation.

Several findings motivate the idea.

Kim, Ganguli & Frank showed that SUB contains distributed spatial representations and theta phase precession, but with spatial coding properties different from CA1. PubMed

Simonnet & Brecht found that subicular principal neurons divide into sparsely versus dominantly bursting firing regimes; dominantly bursting cells tend to have higher firing rates and substantially weaker spatial modulation, whereas sparsely bursting cells carry more detailed spatial information. PubMed

Böhm et al. found that bursting SUB cells are preferentially recruited during hippocampal ripples, while regular-firing cells are suppressed relative to them, suggesting a privileged hippocampal-output channel during replay. PubMed +1

Lee et al. showed that SUB neurons encode combinations of spatial, contextual, and behavioral variables and that theta contributes to organizing these task variables.

PL... +1

Sharp found surprisingly stable SUB spatial representations across very different environments, suggesting that SUB coding can retain common structure where hippocampal representations remap much more strongly. PubMed +1

These findings motivate — but do **not demonstrate** — your proposed decomposition:

$$SUB_i = C_i + P_i.$$

That distinction is important.

References from our discussion

I've grouped these by how directly useful I think they are for the paper.

A. Your work and the new Dembitskaya paper

Baron, L., Diba, K., & Amarasingham, A. (2026). The Role of Plasticity in Replay: Stability Through Anti-Hebbian Rules. *Hippocampus, 36*, e70089. DOI: 10.1002/hipo.70089. Dembitskaya et al. explicitly cite this as their reference 53 when discussing anti-Hebbian plasticity as a possible mechanism for stabilizing hippocampal replay. Dembitskaya et al 2026 - Irregu...

Dembitskaya, Y., Valtcheva, S., Cui, Y., Zheng, Z., Ostojic, S., Graupner, M., & Venance, L. (2026). Irregular and bursty neuronal firing extend spike-timing dependent plasticity window to behavioral timescales. *iScience, 29*, 117052. DOI: 10.1016/j.isci.2026.117052. Their central result is that irregular/bursty firing profoundly changes effective STDP and can make bursts disproportionately responsible for plasticity. Dembitskaya et al 2026 - Irregu...

B. CA3→CA1 / Schaffer collateral plasticity

Wittenberg, G. M., & Wang, S. S.-H. (2006). Malleability of spike-timing-dependent plasticity at the CA3–CA1 synapse. *Journal of Neuroscience, 26*, 6610–6617. DOI: 10.1523/JNEUROSCI.5388-05.2006. This is important because CA3→CA1 plasticity depends strongly on firing pattern/frequency and cannot be described by one immutable pairwise STDP curve. PubMed

Sadowski, J. H. L. P., Jones, M. W., & Mellor, J. R. (2016). Sharp-Wave Ripples Orchestrate the Induction of Synaptic Plasticity during Reactivation of Place Cell Firing

Patterns in the Hippocampus. *Cell Reports, 14*, 1916–1929. DOI: 10.1016/j.celrep.2016.01.061. PubMed

Cepeda-Prado, E. A., et al. (2022). Calcium-Permeable AMPA Receptors Mediate Timing-Dependent LTP Elicited by Low Repeat Coincident Pre- and Postsynaptic Activity at Schaffer Collateral–CA1 Synapses. *Cerebral Cortex, 32*, 1682–1703. DOI: 10.1093/cercor/bhab306. This is discussed explicitly by Dembitskaya et al. in relation to low-repeat/burst-dependent Schaffer-collateral LTP. Dembitskaya et al 2026 - Irregu…

Bi, G.-Q., & Poo, M.-M. (1998). Synaptic modifications in cultured hippocampal neurons: Dependence on spike timing, synaptic strength, and postsynaptic cell type. *Journal of Neuroscience, 18*, 10464–10472. DOI: 10.1523/JNEUROSCI.18-24-10464.1998. Dembitskaya et al 2026 - Irregu…

C. Subiculum firing and spatial/context representation

Böhm, C., et al. (2015). Functional Diversity of Subicular Principal Cells during Hippocampal Ripples. *Journal of Neuroscience, 35*, 13608–13618. DOI: 10.1523/JNEUROSCI.5034-14.2015. Bursting cells are preferentially engaged during ripples and the RS/BS populations have distinct local network connectivity. PubMed +1

Simonnet, J., & Brecht, M. (2019). Burst Firing and Spatial Coding in Subicular Principal Cells. *Journal of Neuroscience, 39*, 3651–3662. DOI: 10.1523/JNEUROSCI.1656-18.2019. Particularly important for your context-backbone hypothesis: dominantly bursting cells tend to carry relatively little precise spatial information. PubMed

Kim, S. M., Ganguli, S., & Frank, L. M. (2012). Spatial Information Outflow from the Hippocampal Circuit: Distributed Spatial Coding and Phase Precession in the Subiculum. *Journal of Neuroscience, 32*, 11539–11558. DOI: 10.1523/JNEUROSCI.5942-11.2012. JN The Journal of … +1

Lee, S.-M., Seol, J.-M., & Lee, I. (2022). Subicular neurons represent multiple variables of a hippocampal-dependent task by using theta rhythm. *PLoS Biology, 20*, e3001546. DOI: 10.1371/journal.pbio.3001546. PL… +1

Sharp, P. E. (1997). Subicular cells generate similar spatial firing patterns in two geometrically and visually distinctive environments: comparison with hippocampal place cells. *Behavioural Brain Research, 85*, 71–92. PubMed

Sharp, P. E. (2006). Subicular place cells generate the same “map” for different environments: Comparison with hippocampal cells. *Behavioural Brain Research, 174*, 206–214. DOI: 10.1016/j.bbr.2006.07.034. PubMed

The 2026 **“Spatial remapping in the subicular complex and entorhinal cortex follows low-dimensional geometric principles”** bioRxiv preprint we discussed is also worth tracking, but because it is a very recent preprint I would treat its claims as preliminary. bioR…

D. CA1→SUB plasticity

Pandey, A., & Sikdar, S. K. (2014). Depression biased non-Hebbian spike-timing-dependent synaptic plasticity in the rat subiculum. *Journal of Physiology*. PMID 24907304. This is the paper relevant to your anti-Hebbian CA1→SUB rule. PubMed +1

Wozny, C., Maier, N., Schmitz, D., & Behr, J. (2008). Two different forms of long-term potentiation at CA1–subiculum synapses. The study showed substantial cell-type dependence of synaptic plasticity in regular-firing versus bursting SUB neurons. PubMed Centr…

Moore, S. J., et al. (2009). Plasticity of burst firing in subiculum induced by synergistic activation of NMDA receptors and calcium channels. This is relevant if you eventually want intrinsic burst propensity itself to be plastic rather than fixed. PubMed Centr…

E. Entorhinal deep↔superficial circuitry

Ohara, S., et al. (2018). Intrinsic Projections of Layer Vb Neurons to Layers Va, III, and II in the Lateral and Medial Entorhinal Cortex of the Rat. *Cell Reports, 24*, 107–116. This is probably the most useful anatomical reference for the **deep→superficial EC feedback loop**. ScienceDir…

Ohara, S., et al. (2021). Local projections of layer Vb-to-Va are more prominent in lateral than medial entorhinal cortex. *eLife*. They show that LVb also preferentially targets superficial layers III/II, with meaningful MEC/LEC differences. eL…

Simonsen, Ø. W., et al. (2022). Retrosplenial and subicular inputs converge on superficially projecting neurons in medial entorhinal cortex. This is particularly relevant because it links **SUB input to deep EC neurons that themselves project superficially**. Springer Li…

Ohara, S., et al. (2023). Hippocampal–medial entorhinal circuit is differently organized along the dorsoventral axis in rodents. *Cell Reports*. Among other things, it refines which deep EC populations receive dorsal versus ventral hippocampal output. C…

Maass, A., et al. (2014). Laminar activity in the hippocampus and entorhinal cortex related to novelty and episodic encoding. *Nature Communications, 5*, 5547. DOI: 10.1038/ncomms6547. This provides human evidence consistent with the broad superficial-input/deep-output organization and associates deep EC activity with subsequent memory. Nature +1

F. Predictive hippocampal/sequence models

Chen, Y., Zhang, H., Cameron, M., & Sejnowski, T. J. (2024). Predictive sequence learning in the hippocampal formation. *Neuron, 112*, 2645–2658.e4. DOI: 10.1016/j.neuron.2024.05.024. This is probably one of the most important computational references for your proposed extension because it explicitly formulates hippocampal sequence processing as prediction. PubMed

Bouhadjar, Y., Wouters, D. J., Diesmann, M., & Tetzlaff, T. (2022). Sequence learning, prediction, and replay in networks of spiking neurons. This work shows that spiking networks with local Hebbian/homeostatic mechanisms can perform **context-specific next-element prediction** and autonomous replay. ar...

Spens, E., & Burgess, N. (2024). A generative model of memory construction and consolidation. Their model explicitly considers hippocampal replay as training slower generative representations, which is highly relevant to your broader "memory export" concept. PubMed

Liao, Z., et al. (2024). Inhibitory plasticity supports replay generalization in the hippocampus. *Nature Neuroscience, 27*, 1987–1998. DOI: 10.1038/s41593-024-01745-w. Important for the idea that consolidation/replay may extract structure rather than preserve every sensory feature indiscriminately. PubMed

Buzsáki, G. (2015). Hippocampal sharp wave-ripple: A cognitive biomarker for episodic memory and planning. *Hippocampus, 25*, 1073–1188. A particularly relevant conceptual point is the idea that SWRs help transfer compressed hippocampal representations to distributed circuits. PubMed

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The ICLR narrative I would currently use

If the next experiments work, I think the paper's conceptual progression should become:

1. Fast episodic learning is inherently capacity limited

$$\text{sequential CA3 learning} \rightarrow \text{synaptic saturation} \rightarrow \text{interference.}$$

2. Replay-driven compression improves the cache

$$\text{replay} + \text{synaptic reorganization} \rightarrow \text{greater CA3 efficiency}.$$

But capacity remains finite.

3. Replay performs systems-level consolidation

$$CA3 \xrightarrow{\text{replay}} CA1 \rightarrow SUB \rightarrow EC_{deep}.$$

4. Downstream memory changes representation

Instead of storing another recurrent copy:

$$M_i = (P_1, \dots, P_T),$$

EC learns:

$$P(P_{t+1} \mid P_t, C_i).$$

5. Context prevents destructive interference

$$(C_A, P_t)$$

and

$$(C_B, P_t)$$

can predict different successors even when trajectories overlap.

6. Deep↔superficial EC enables generative recall

$$\text{cue} \rightarrow \text{prediction} \rightarrow \text{reinstated state} \rightarrow \text{next prediction}.$$

7. Successful consolidation allows CA3 eviction

$$L_{\text{prediction}} < \epsilon \Rightarrow \text{release CA3 resources}.$$

8. Long-term capacity is therefore no longer determined by CA3 capacity

That gives you the central claim:

> **Replay does not merely protect memories inside a finite hippocampal network; it can transform transient episodic traces into context-conditioned predictive representations downstream, allowing the finite hippocampal store to be continually recycled.**

For an ICLR audience, I would make the analogy explicit but not gimmicky:

$$\boxed{\text{CA3} = \text{fast finite memory/cache}}$$

$$\boxed{\text{replay} = \text{consolidation/training process}}$$

$$\boxed{\text{EC} = \text{slower predictive model}}$$

$$\boxed{\text{prediction error} = \text{consolidation/eviction criterion.}}$$

If you can demonstrate **old-memory cued recall after the corresponding CA3 trace has been removed and dozens of later environments have been learned**, I think that would become the most consequential result in the paper. It turns your current finding—that compression only delays interference—into the motivation for a fundamentally different solution to continual memory.